

PV–PWR–4 power board

Rev A2 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PV–P100/110: 4 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

01 PC connector, command buffers, status

Interfaces PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points

File: sheets/01_pc.kicad_sch

02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/02_ph1_power.kicad_sch

03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03_ph1_gda.kicad_sch

04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04_ph1_gdb.kicad_sch

05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/05_ph2_power.kicad_sch

06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06_ph2_gda.kicad_sch

07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07_ph2_gdb.kicad_sch

08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/08_ph3_power.kicad_sch

09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09_ph3_gda.kicad_sch

10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10_ph3_gdb.kicad_sch

11 Phase 4 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/11_ph4_power.kicad_sch

12 Phase 4 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/12_ph4_gda.kicad_sch

13 Phase 4 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/13_ph4_gdb.kicad_sch

14 Port A power path (pv)

Varistor Y + monitor

File: sheets/14_leanA_power.kicad_sch

15 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/15_leanA_ctrl.kicad_sch

16 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/16_leanA_drive.kicad_sch

16 Port B power path (battery)

CAUTION: not for use as a design_check() asserts every number on each build (outputs: PV–PWR–4 design_check.txt), drafted @ few days from design, 06/09/2026. Domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain), T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX–T1 reinforced winding -> J_SELV). Coat to PD1. Shared heatsink: Tj 110 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 23.7 W (economiser) <= aux live rating 30.0 W; module aux 82 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

References: PV–PWR–4 design_check.txt, drafted @ few days from design, 06/09/2026. polarity and dV interlocks

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/18_leanB_drive.kicad_sch

18 Terminals, EMI filter, insulation test

DC terminals, nanocrystalline CM ring per port, X / Y capacitors, board entries (press–fit), PE links, heatsink, insulation monitor (port.lean_imd)

File: sheets/18_terminals_imd.kicad_sch

19 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/19_live5v_selv.kicad_sch

20 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–: 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uf / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/20_aux75_input.kicad_sch

21 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms

File: sheets/21_aux75_start.kicad_sch

22 AUX75 PWM, SIC, clamp, AUX–T1

UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 50:9:10; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/22_aux75_pwm.kicad_sch

Not bench–validated. Values read from sim/out/*.json; checks: design_check()

Sheet 1

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

Sheet: 1

File: F

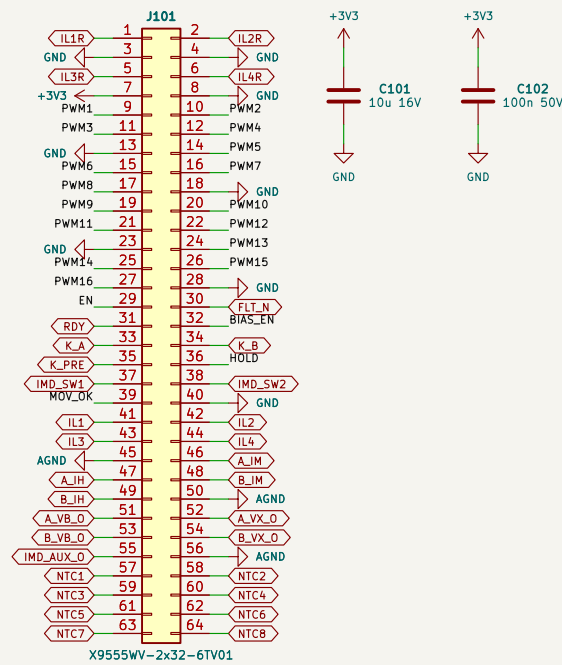
Title

Size: 1

KiCad

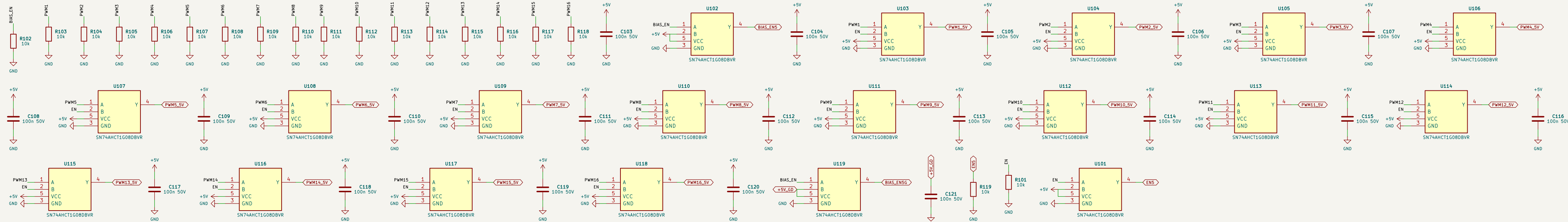
6

PC CONNECTOR (INTERFACES.PC), LIVE



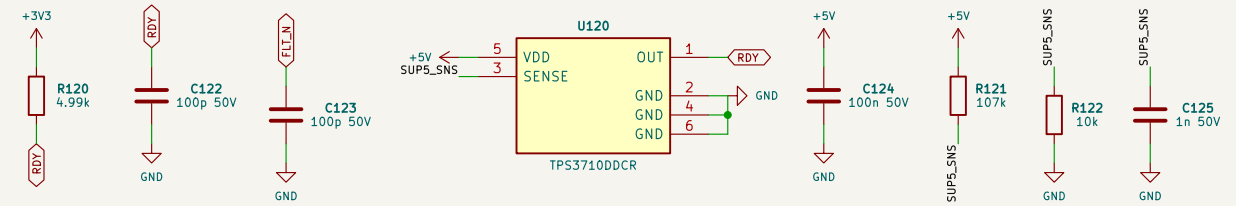
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2: the former +24V / +5V pins carry the sensor references (ILR-ILM)). Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



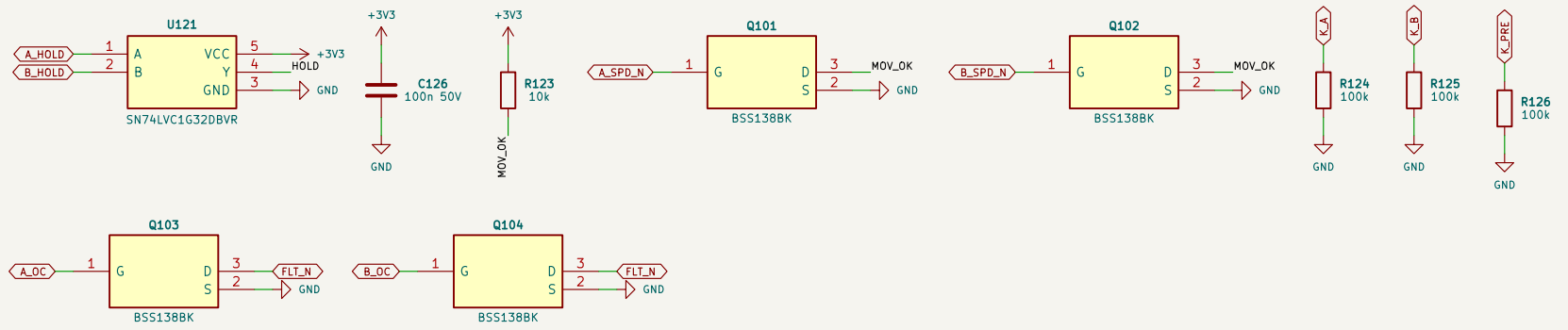
NSI6651 inputs are specified at VCC1 = 5 V only (VIN1 <= 3.3 V, τ = 7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 μ s).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



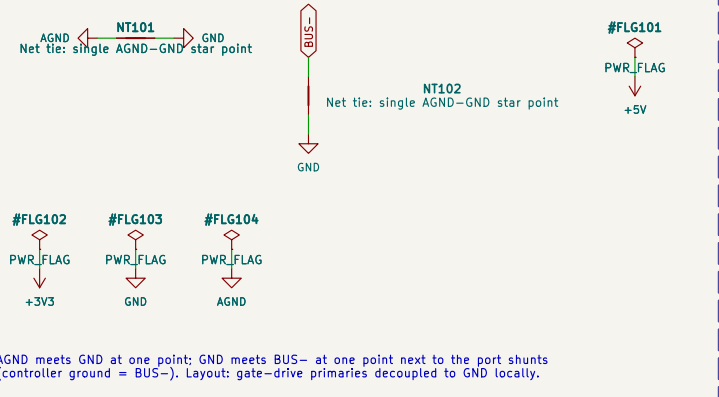
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC (I_{IL} >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K.A / K.B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N

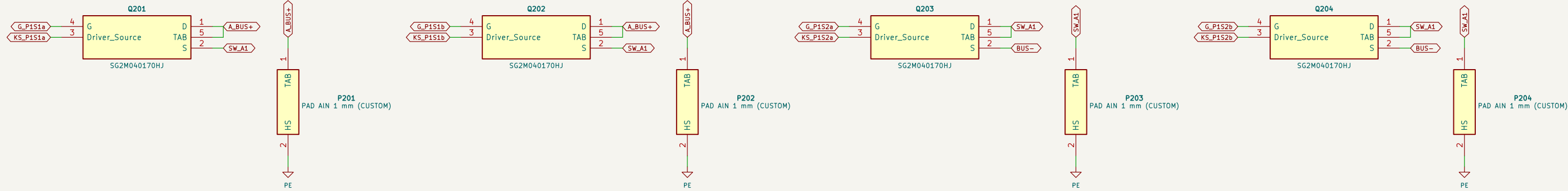


HOLD = A_HOLD OR B_HOLD (|I_{port}| >= 0.97–1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC (I_{IL} >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K.A / K.B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS

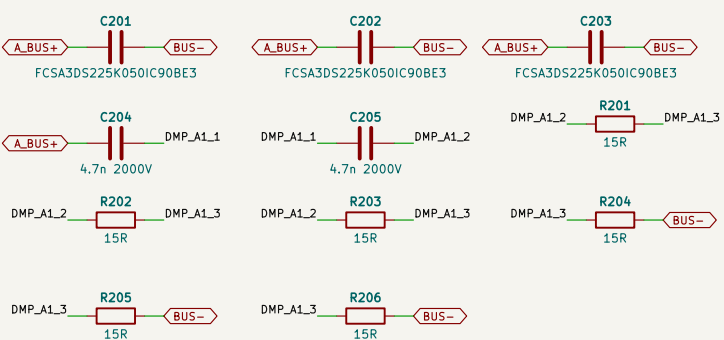


PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



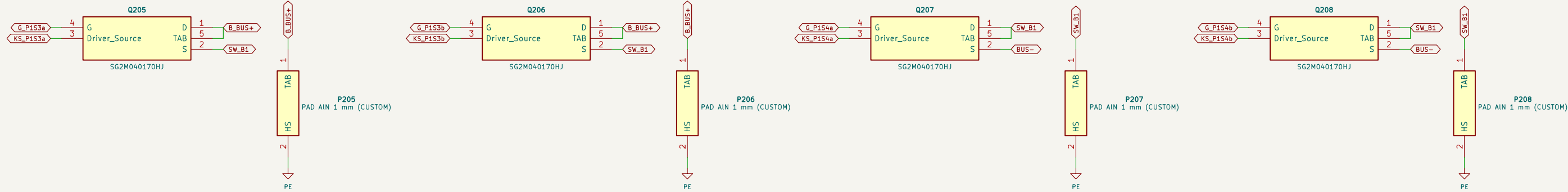
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



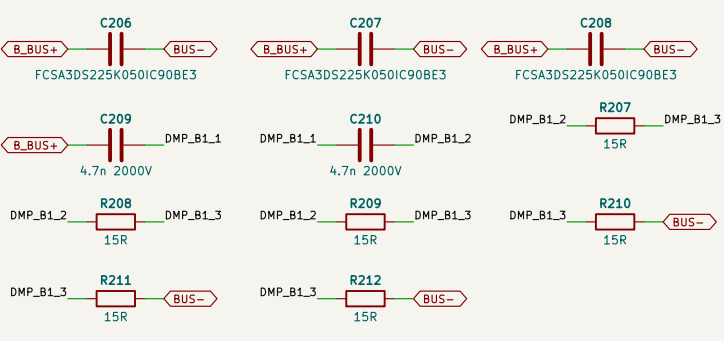
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



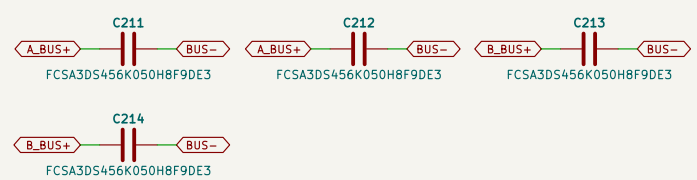
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



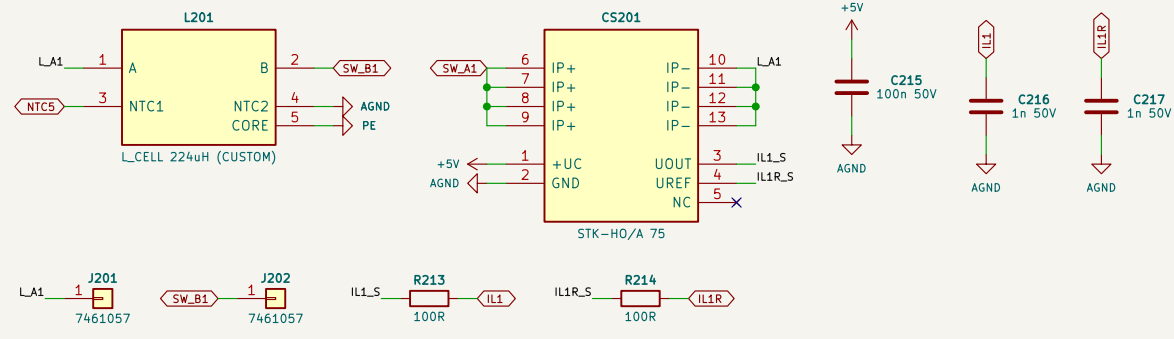
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



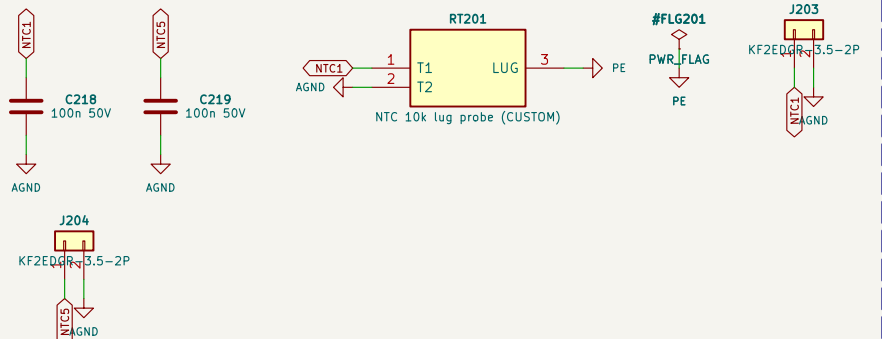
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



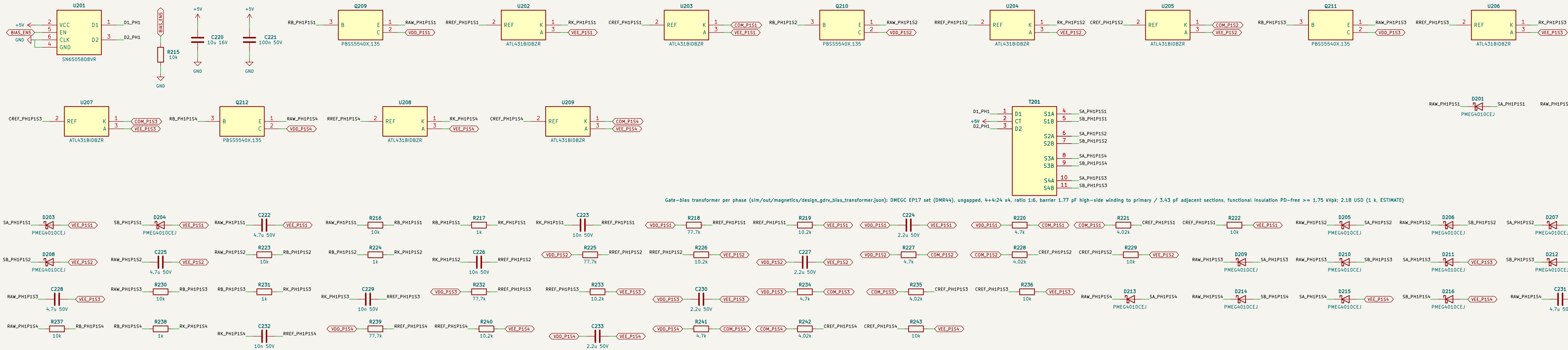
Lead A from SW_A1 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out + = SW_A -> SW_B) to stud
LA1: lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

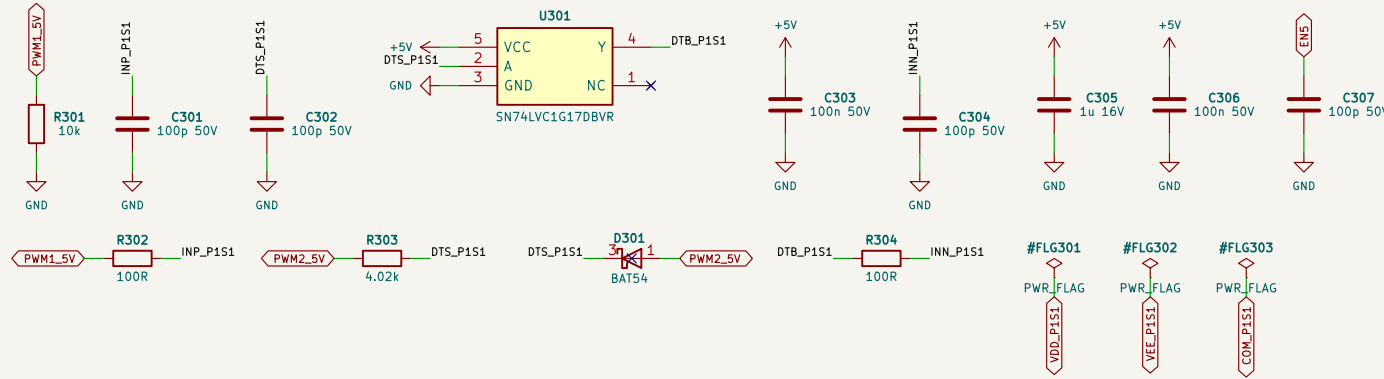
PH1: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMCC EP17 set (DMR44), ungapped, 4+4:2k v4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 U0V (1 k, ESTIMATE)

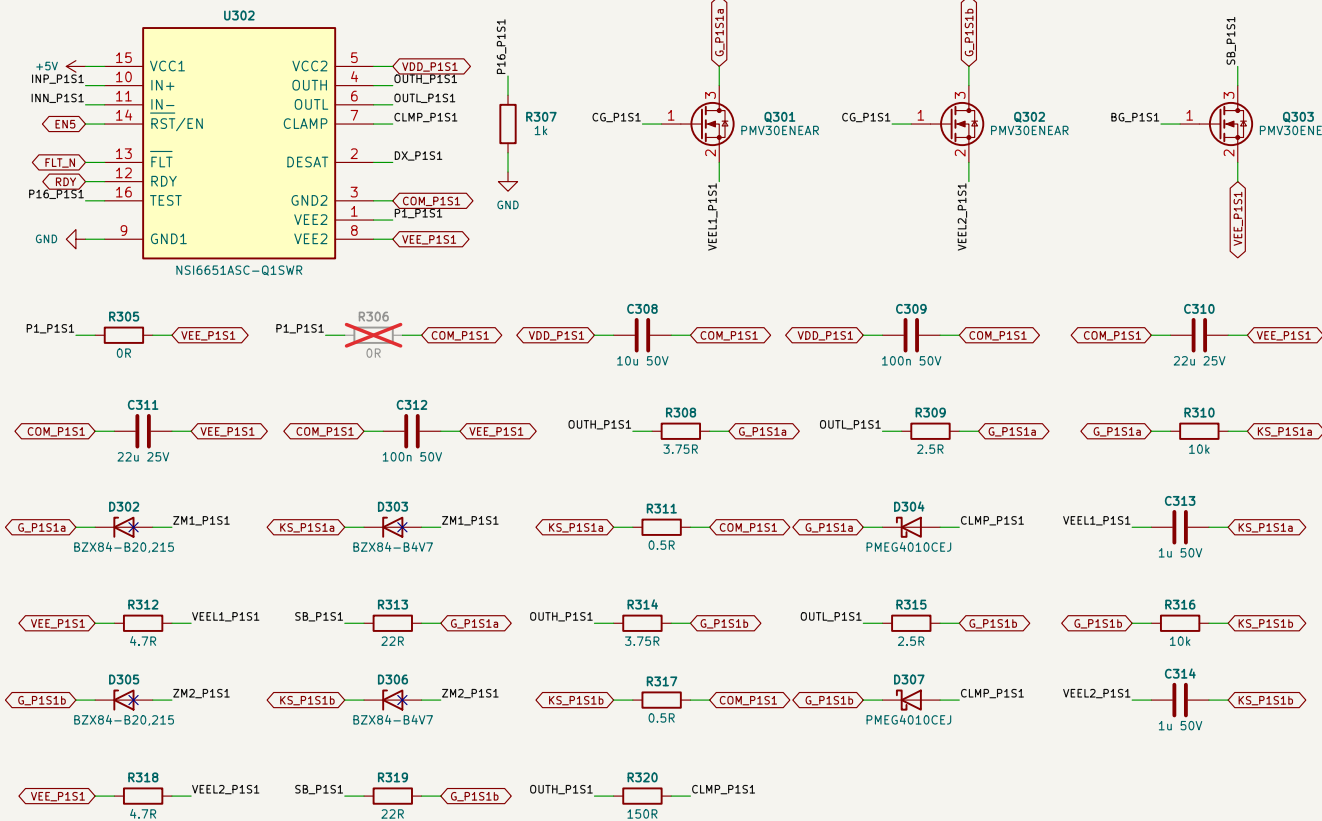
SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S1: LOGIC SIDE



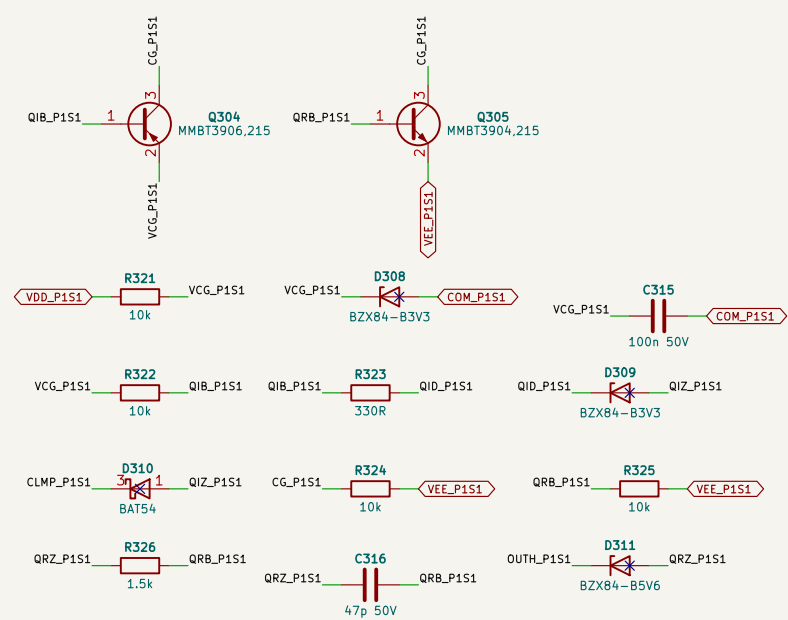
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S1: DRIVER AND 2 GATES



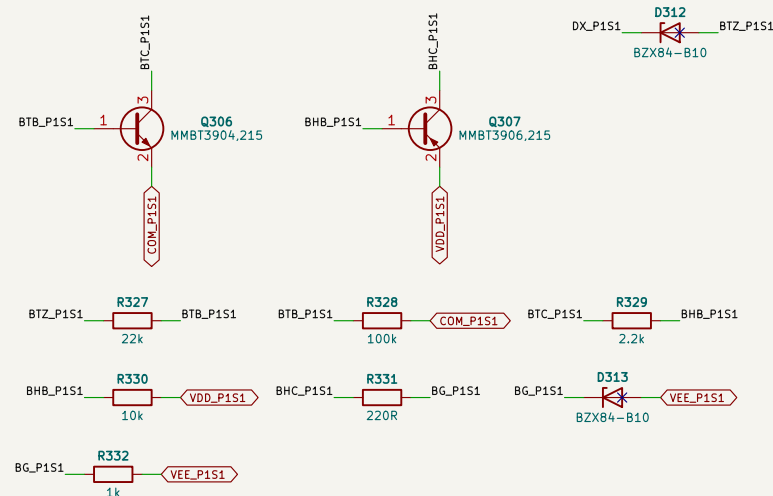
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S1: MILLER CLAMP CONTROL



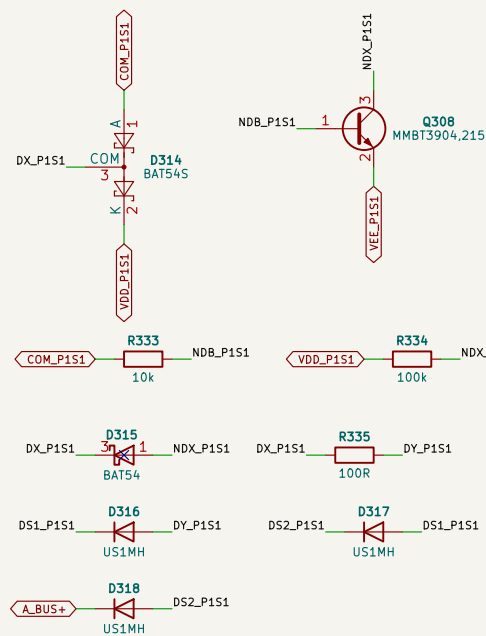
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S1: SHORT-CIRCUIT BOOSTER



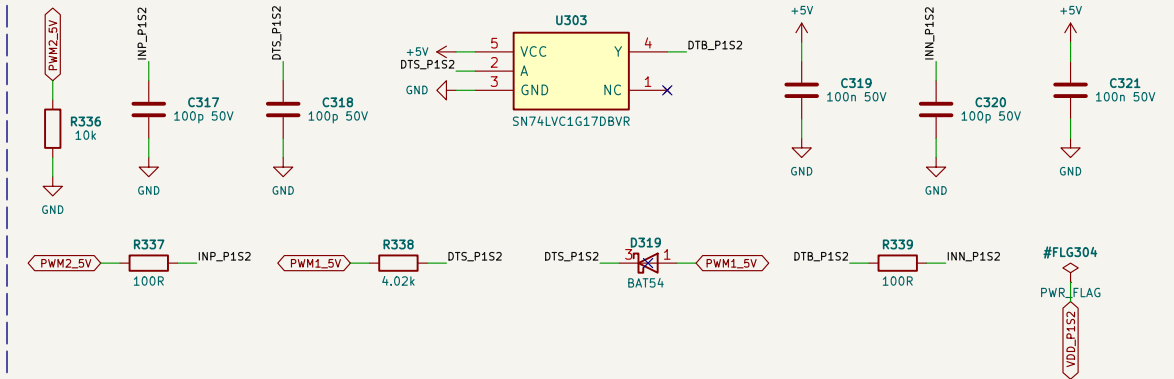
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P1S1: DESAT



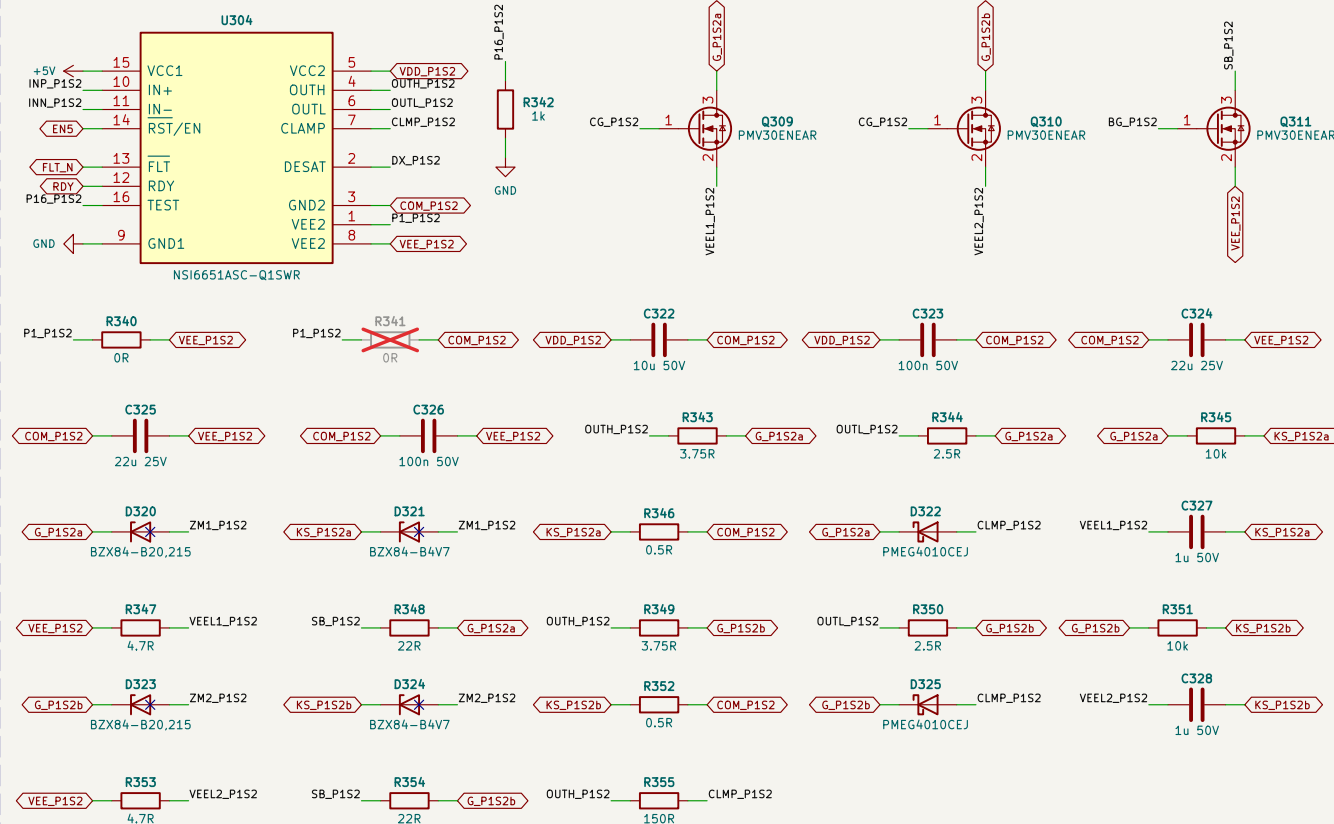
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - R_s = 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S2: LOGIC SIDE



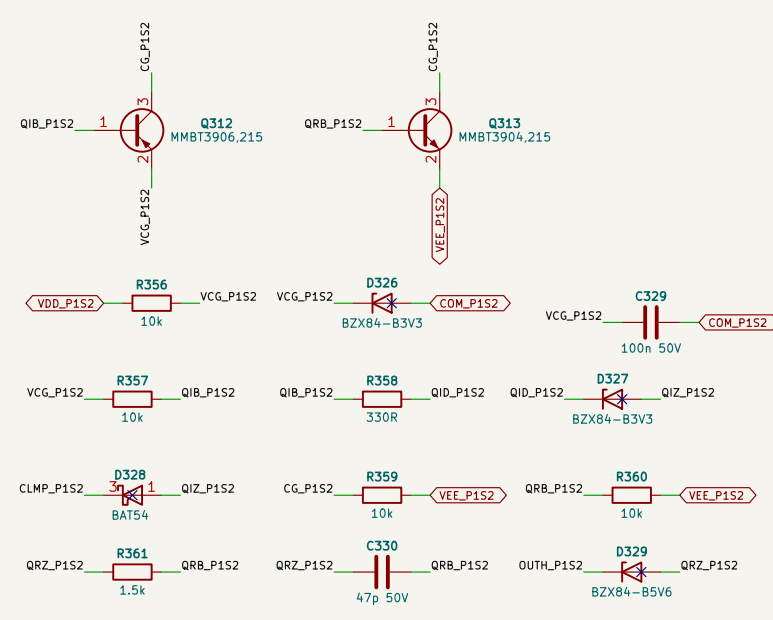
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S2: DRIVER AND 2 GATES



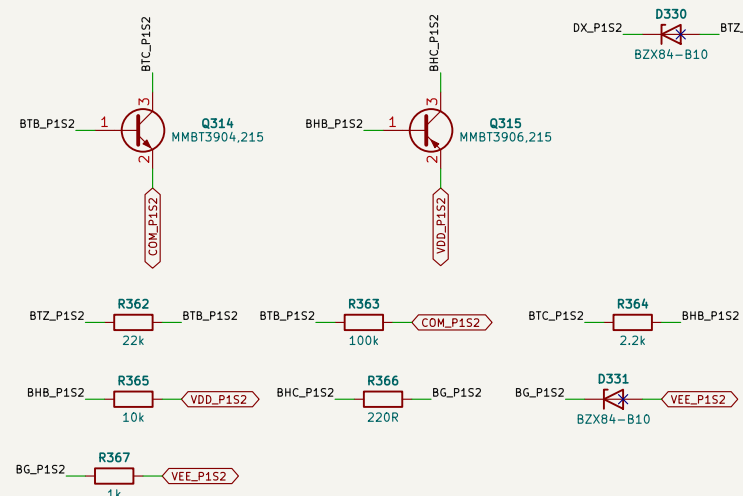
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S2: MILLER CLAMP CONTROL



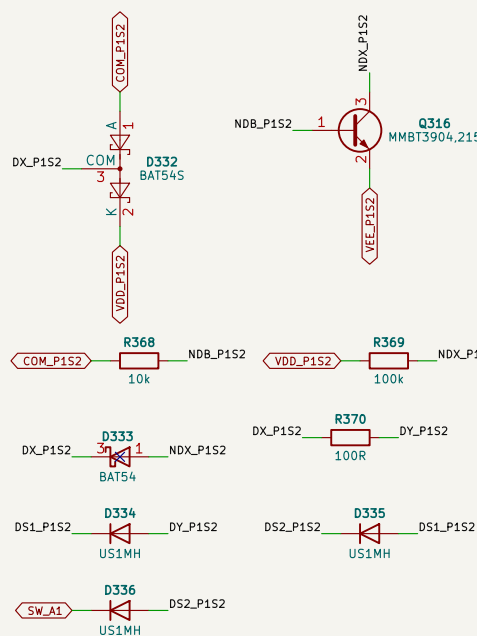
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P1S2: DESAT

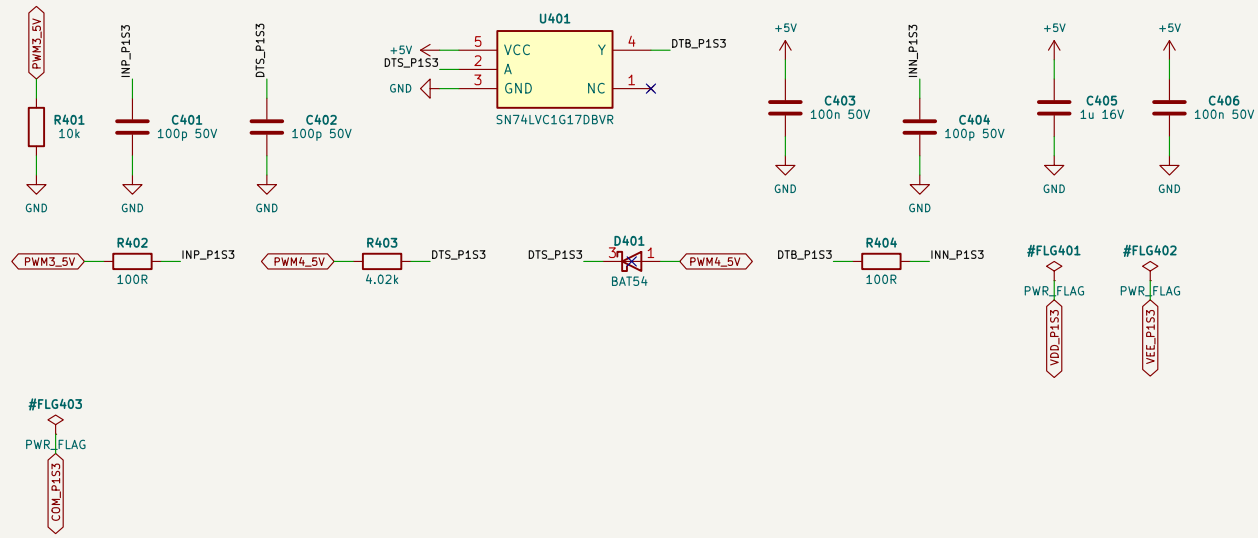


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) - R_s = 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG A: LAYOUT RULES

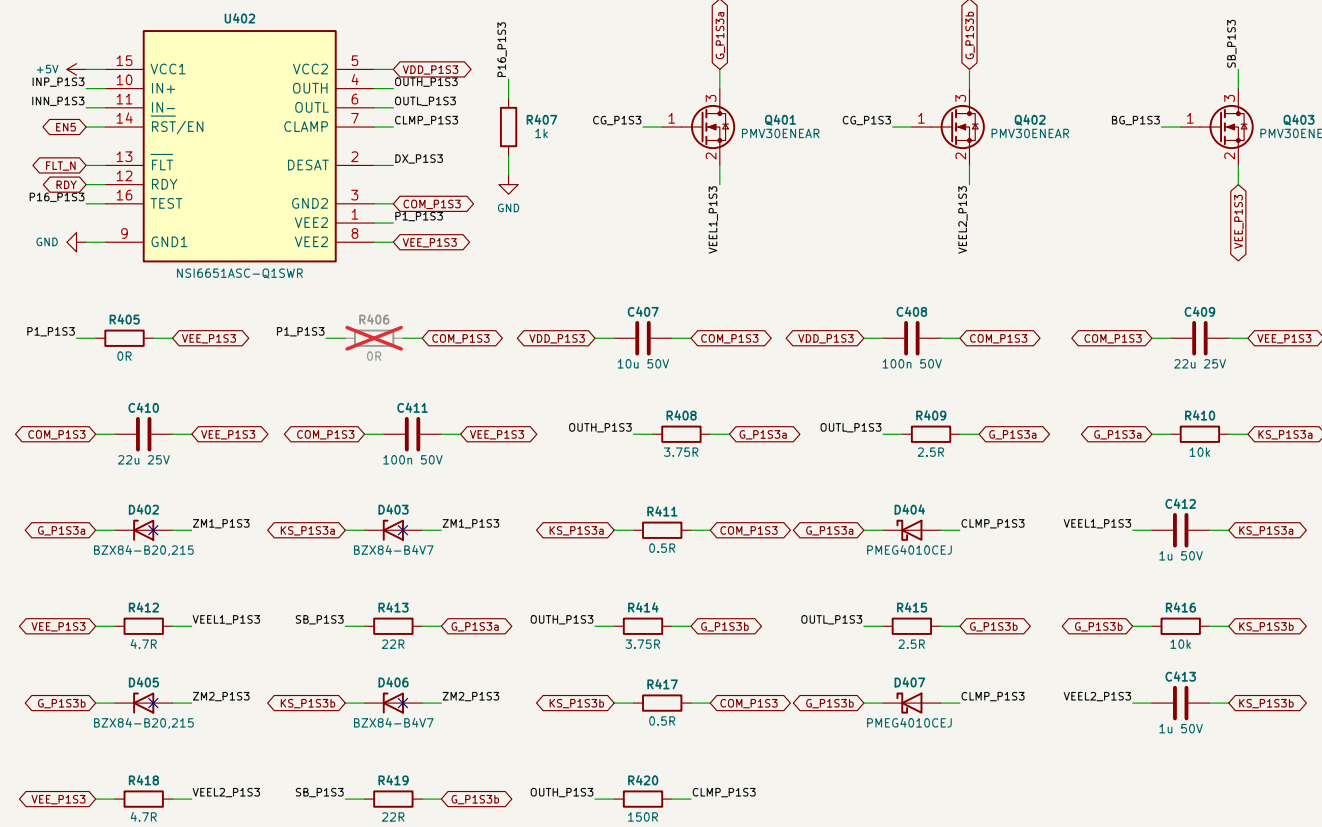
Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{Vt} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nigspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

P1S3: LOGIC SIDE



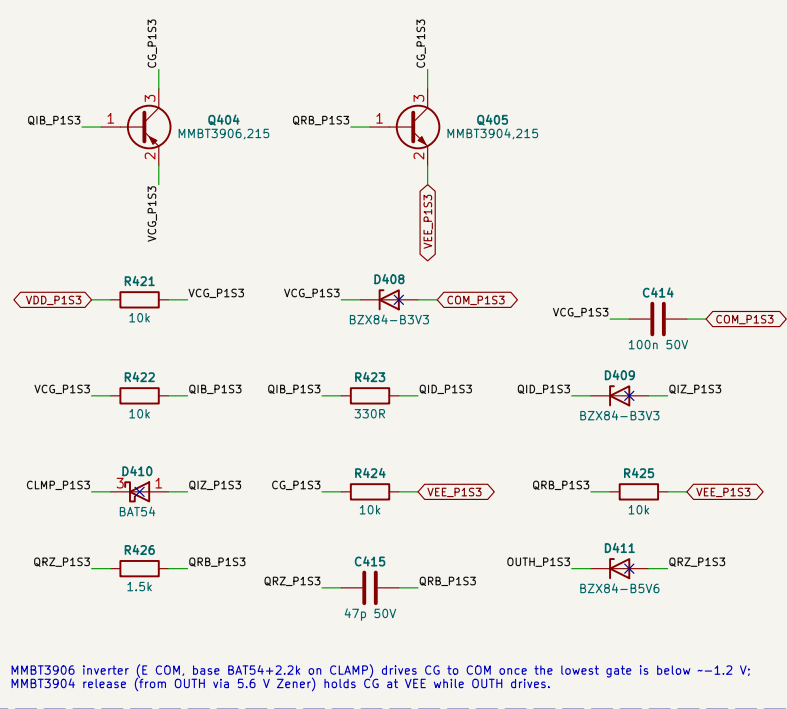
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



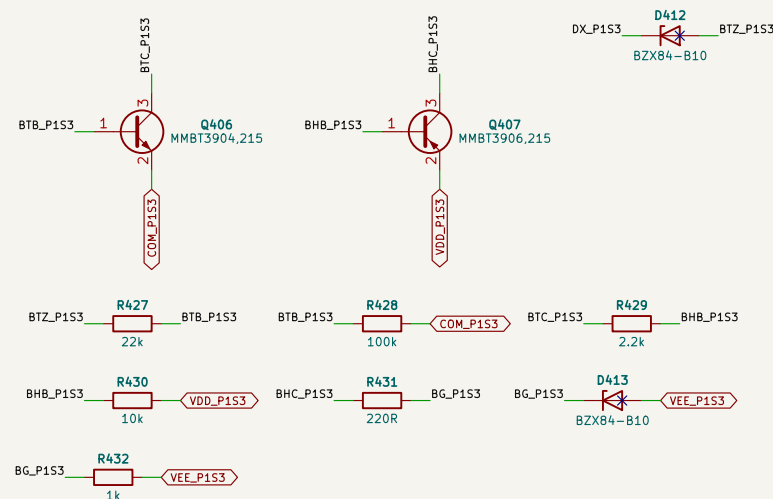
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30EENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



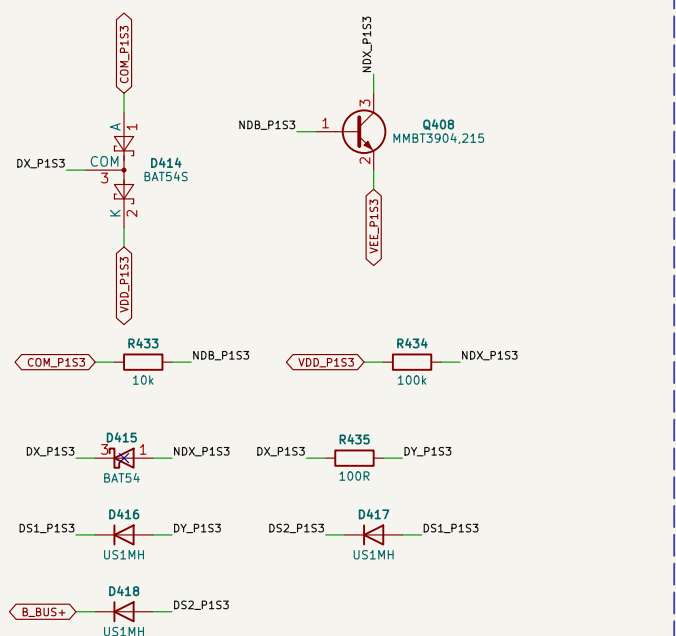
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER



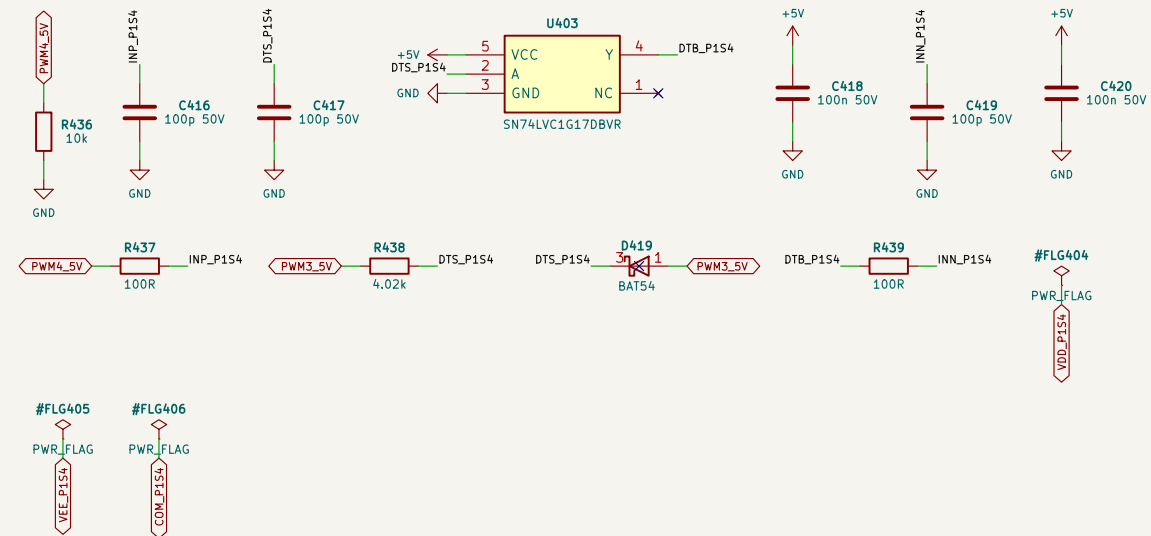
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S3: DESAT



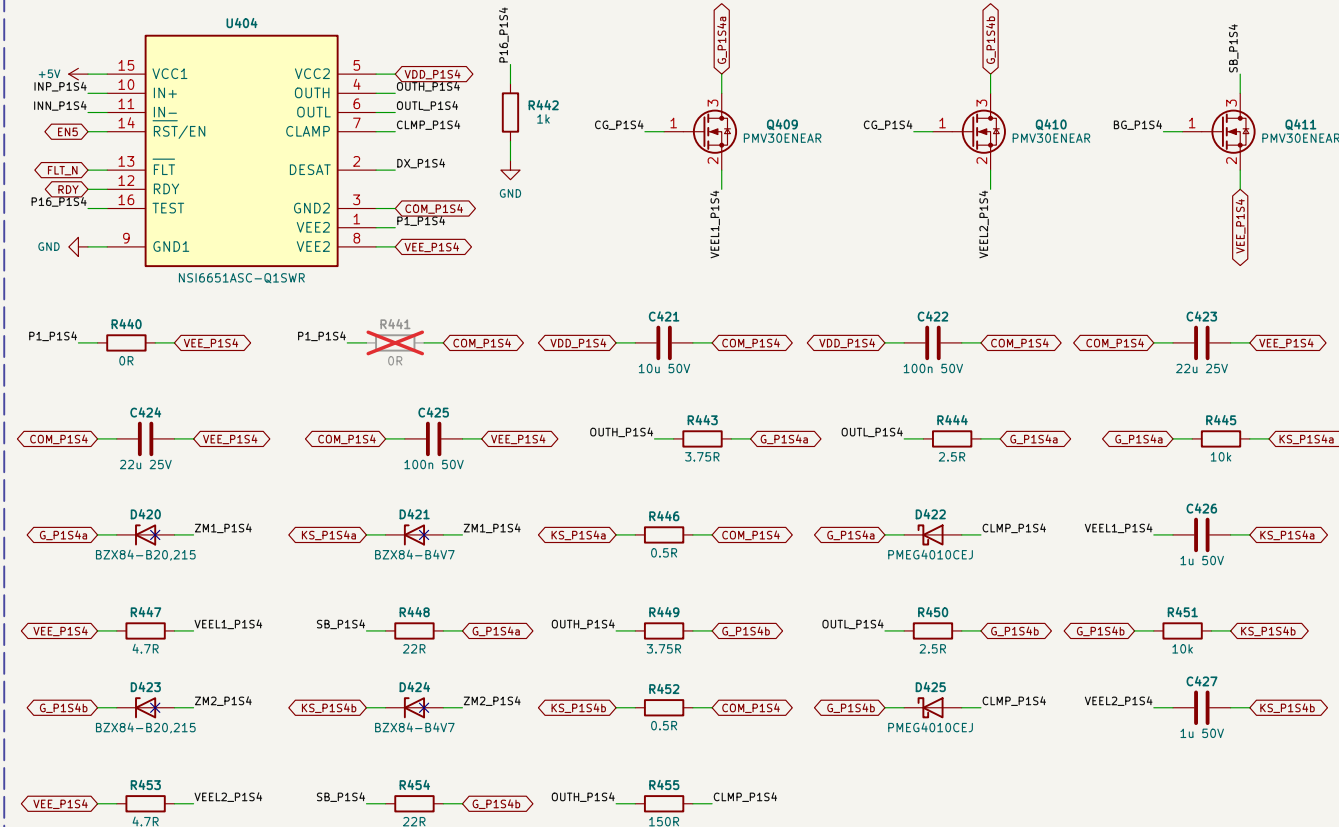
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P1S4: LOGIC SIDE



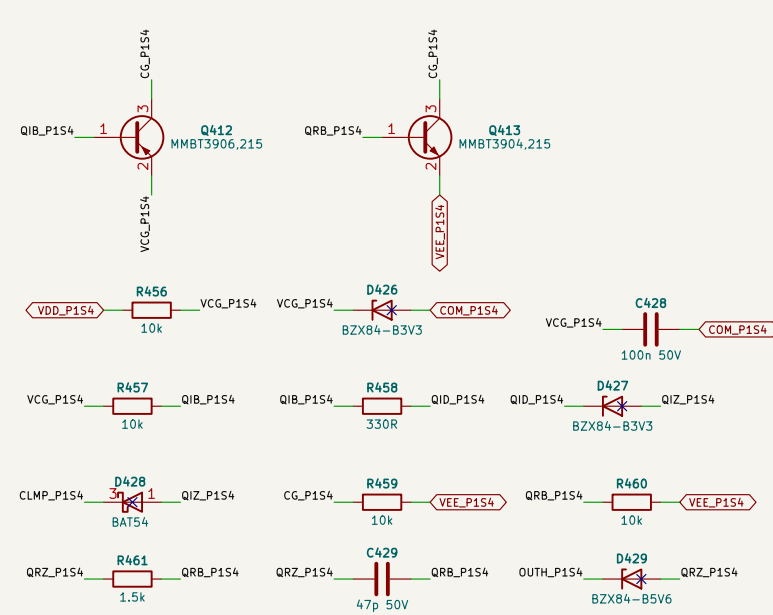
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S4: DRIVER AND 2 GATES



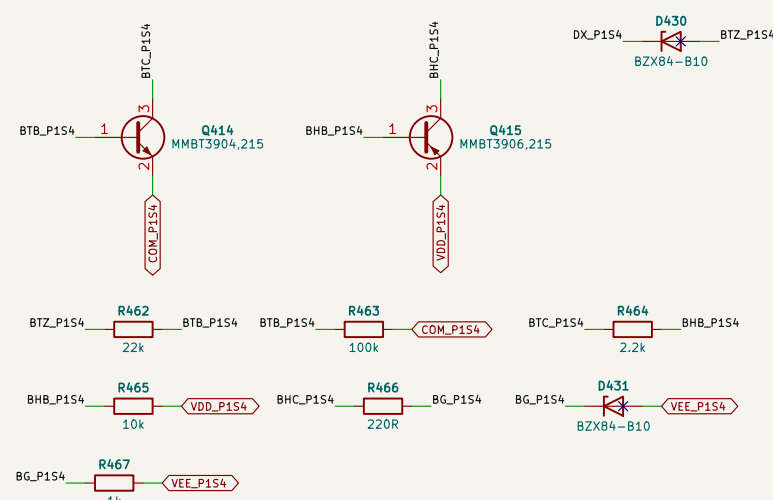
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30EENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S4: MILLER CLAMP CONTROL



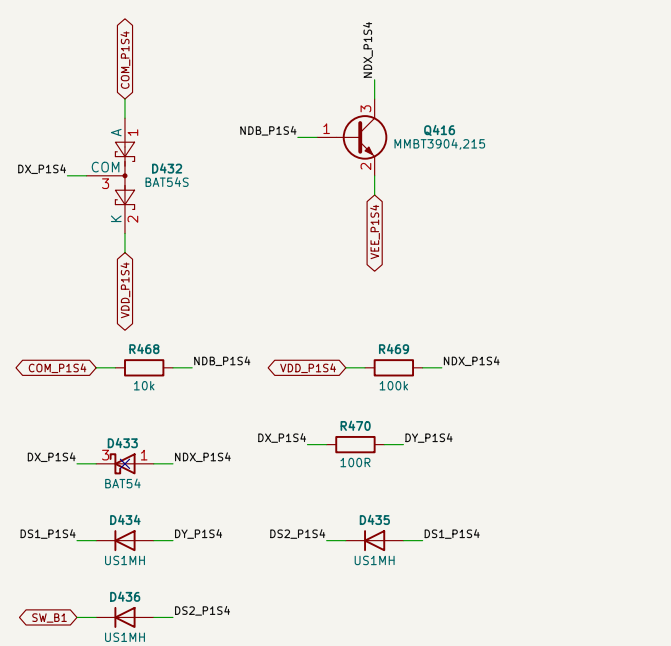
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P1S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nspice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5 - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH

The schematic diagram illustrates the internal components and pin connections for two modules, L501 and C5501.

L501 Module:

- Pin 1 (LA2):** Connected to the A pin of NTC1.
- Pin 2 (SW_B2):** Connected to the B pin of NTC1.
- Pin 3 (NTC6):** Connected to the NTC1 component.
- Pin 4:** Connected to the NTC2 CORE component.
- Pin 5 (AGND PE):** Connected to the AGND pin of NTC2 CORE.
- Internal Component:** LCELL 224uH (CUSTOM) inductor.

C5501 Module:

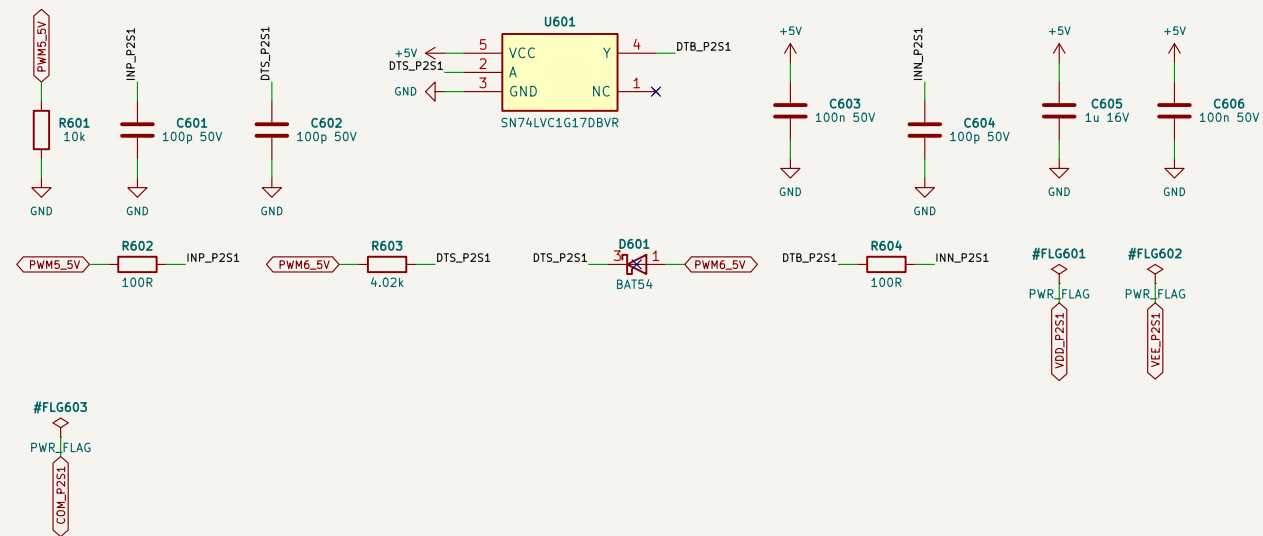
- Pin 6 (SW_A2):** Connected to the IP+ pin of UIC.
- Pin 7:** Connected to the IP+ pin of UIC.
- Pin 8:** Connected to the IP+ pin of UIC.
- Pin 9:** Connected to the IP+ pin of UIC.
- Pin 10 (LA2):** Connected to the IP- pin of UIC.
- Pin 11:** Connected to the IP- pin of UIC.
- Pin 12:** Connected to the IP- pin of UIC.
- Pin 13:** Connected to the IP- pin of UIC.
- Pin 14 (UIC GND):** Connected to the GND pin of UIC.
- Pin 15 (UIC UREF):** Connected to the UREF pin of UIC.
- Pin 16 (UIC NC):** Connected to the NC pin of UIC.
- Internal Components:** UIC, UREF, and UIC components.

External Connections:

- LA2:** Connected to the LA2 pin of the L501 module.
- SW_B2:** Connected to the SW_B2 pin of the L501 module.
- SW_A2:** Connected to the SW_A2 pin of the C5501 module.
- LA2:** Connected to the LA2 pin of the C5501 module.
- UIC GND:** Connected to the UIC GND pin of the C5501 module.
- UIC UREF:** Connected to the UIC UREF pin of the C5501 module.
- UIC NC:** Connected to the UIC NC pin of the C5501 module.
- AGND:** Connected to the AGND pin of the L501 module.
- AGND:** Connected to the AGND pin of the C5501 module.
- AGND:** Connected to the AGND pin of the C5501 module.
- AGND:** Connected to the AGND pin of the C5501 module.
- AGND:** Connected to the AGND pin of the C5501 module.

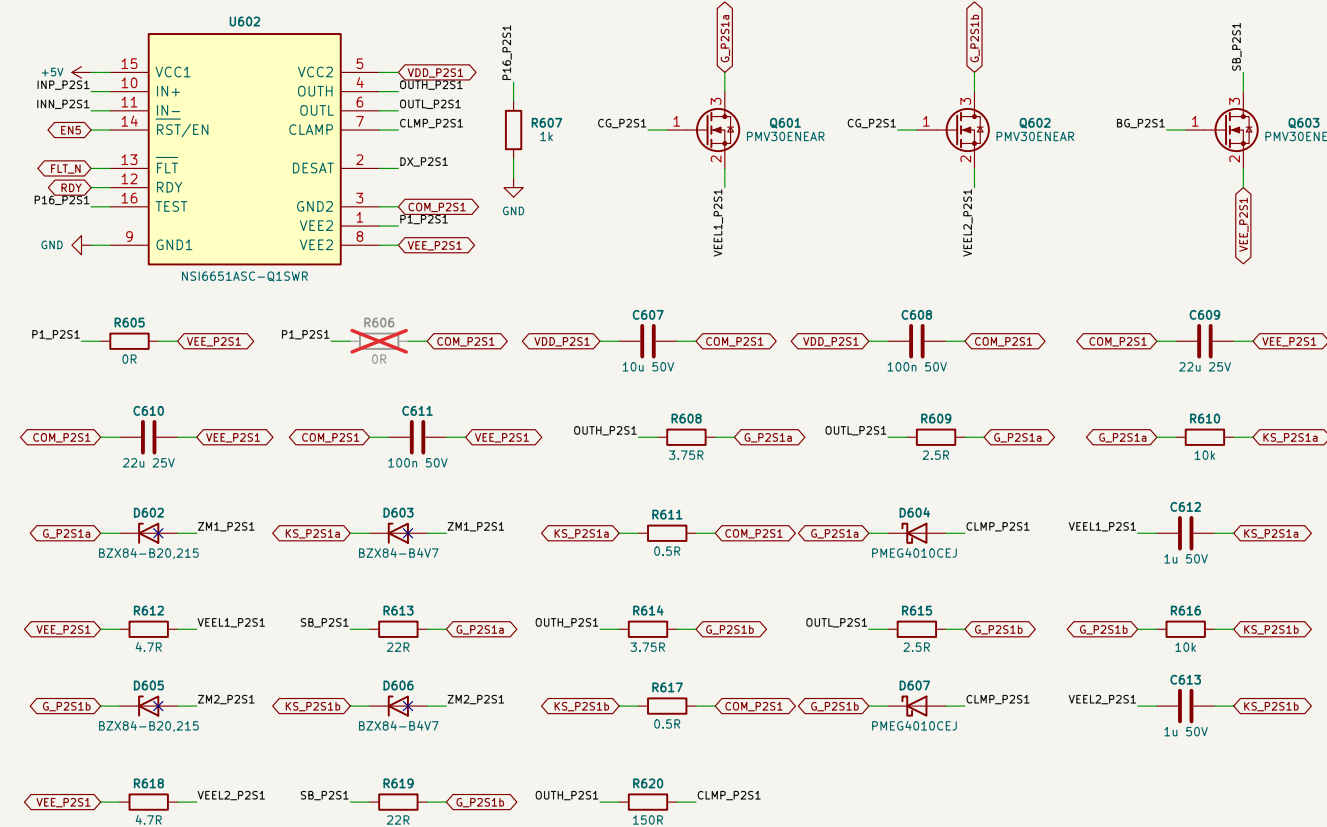
[illegible][illegible]

P2S1: LOGIC SIDE



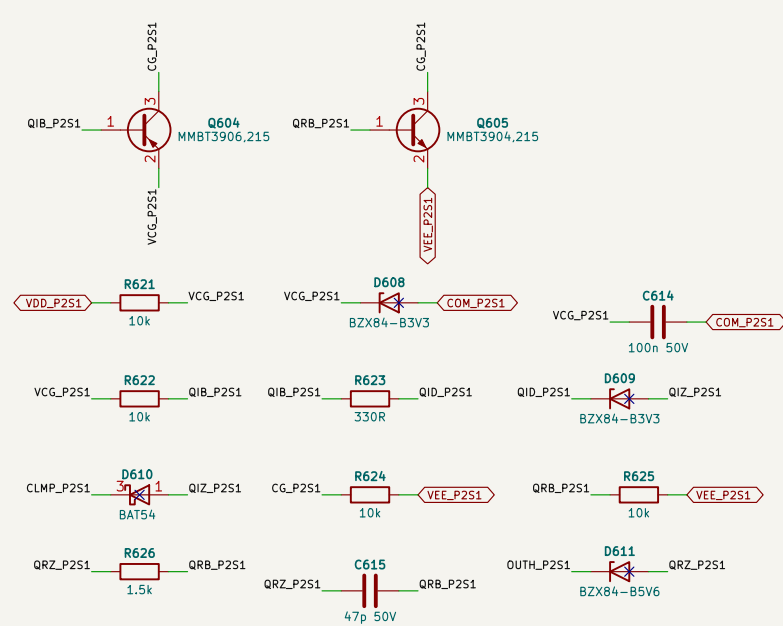
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



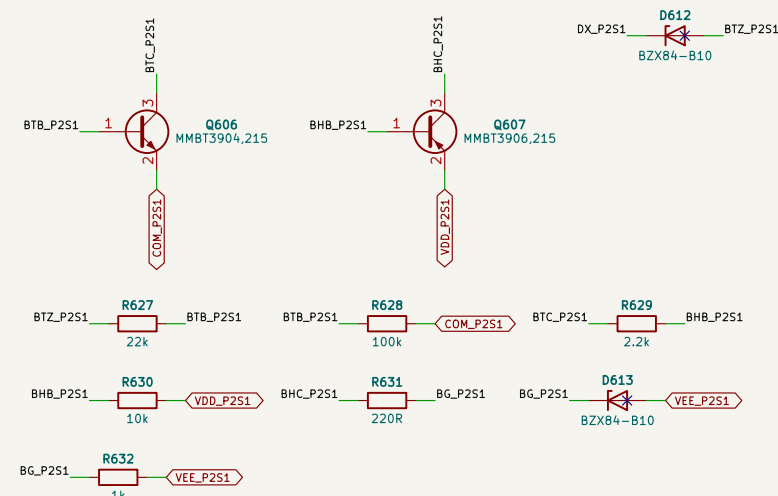
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL



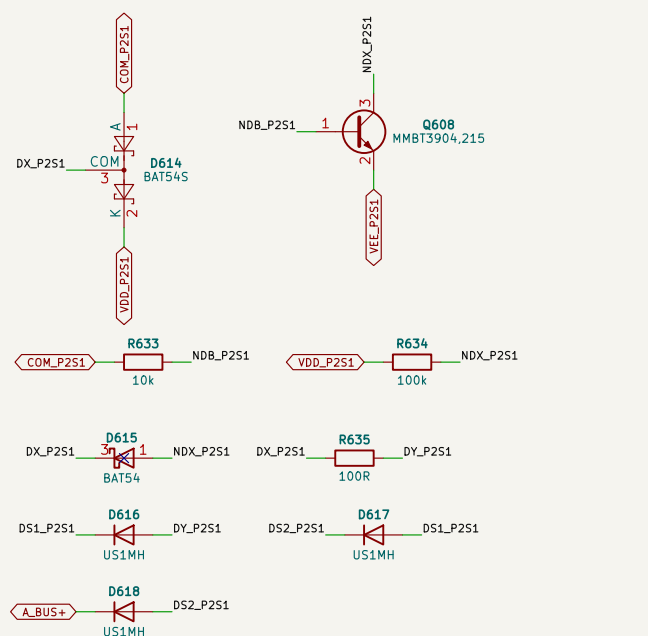
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER



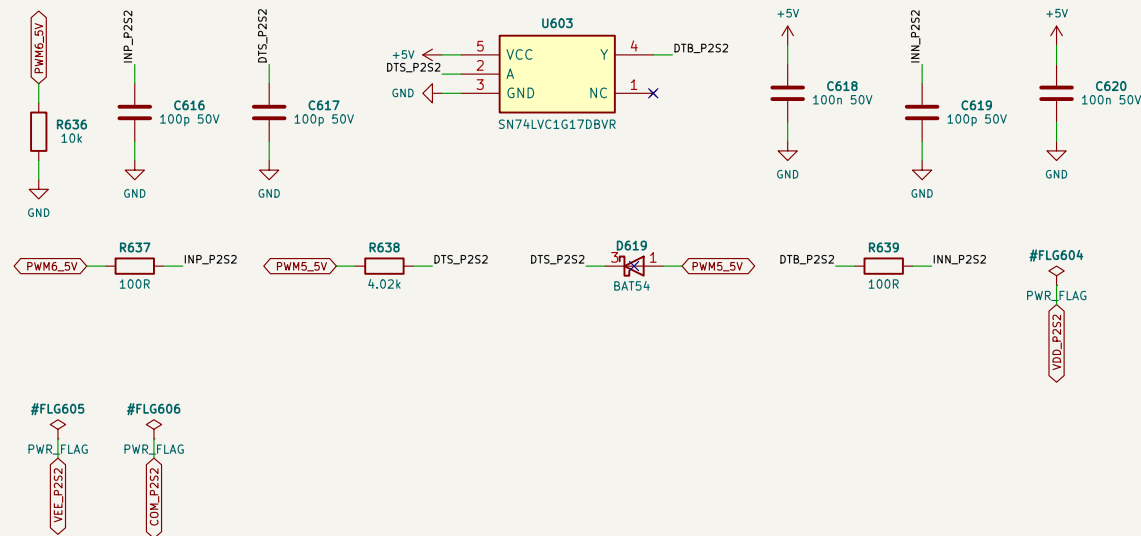
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P2S1: DESAT



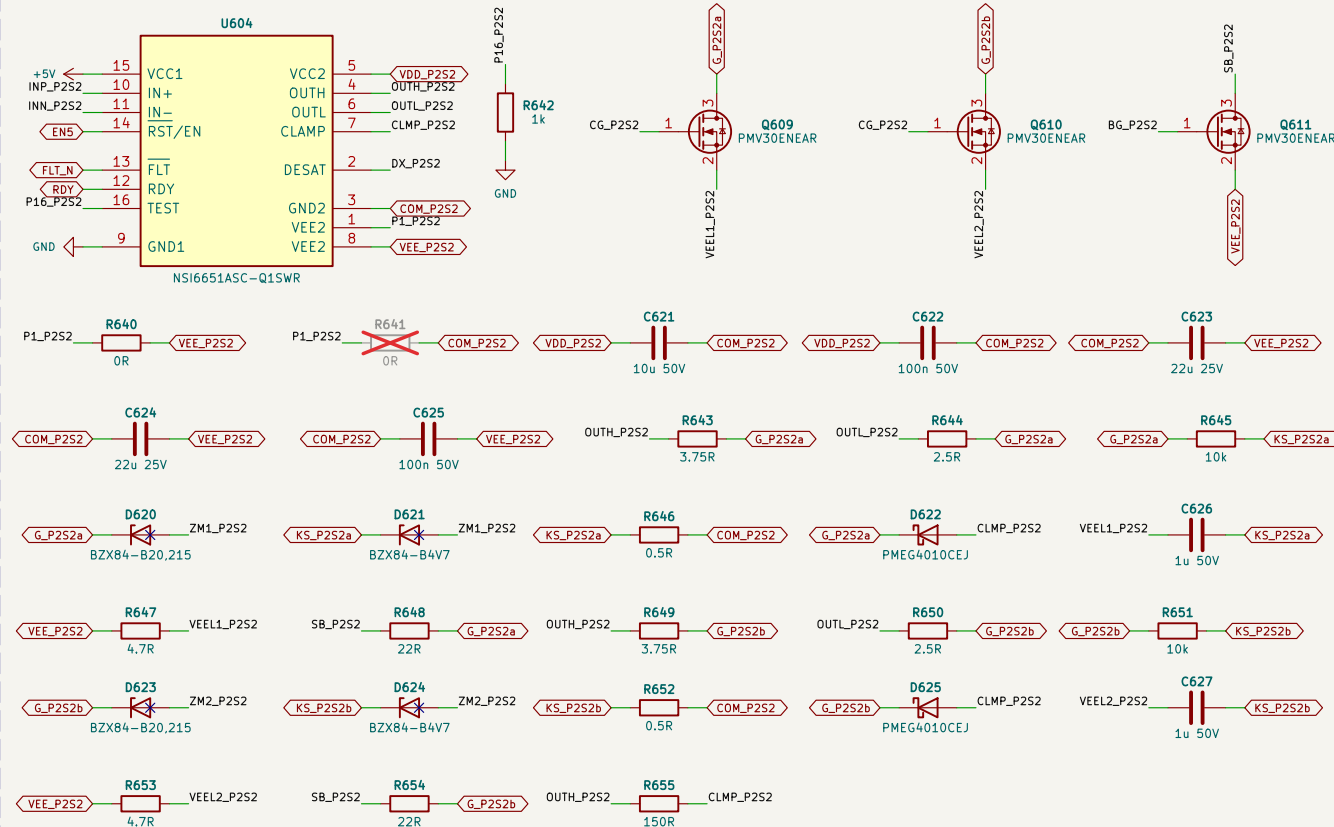
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string: Cblk footprint DNP.

P2S2: LOGIC SIDE



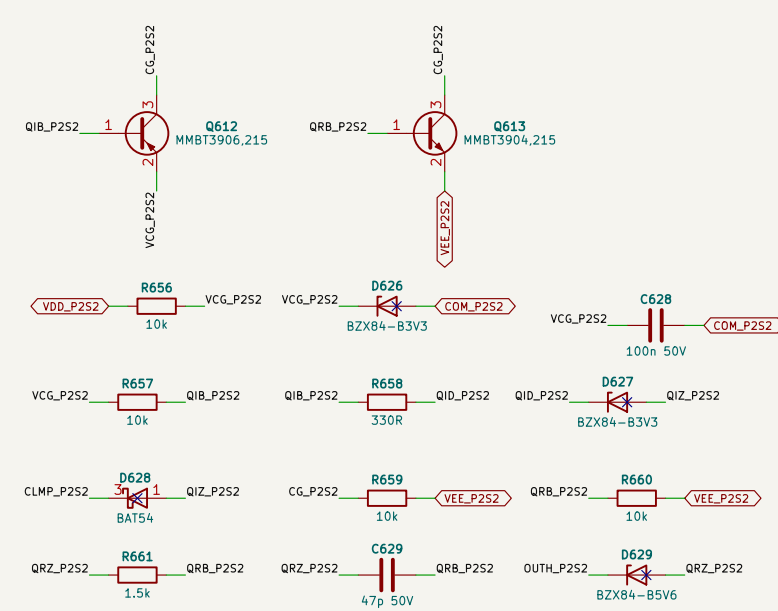
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S2: DRIVER AND 2 GATES



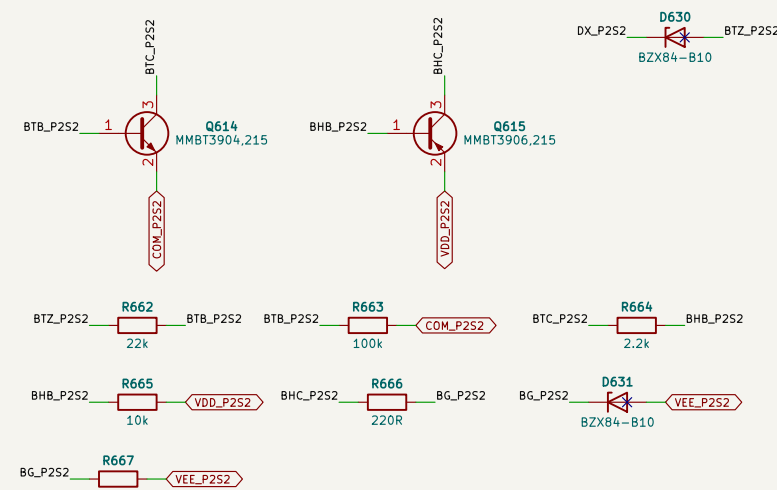
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S2: MILLER CLAMP CONTROL



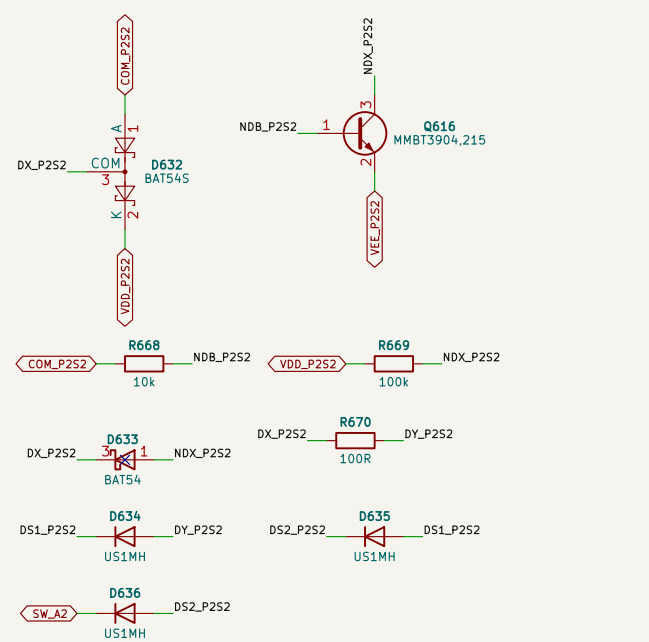
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P2S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 3 x US1M – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k. Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 °C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 7
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basi

Sheet: /06 Phase 2 leg A gate drive (S1, S2)/
File: 06_ph2_gda.kicad.sch

Title: PV-PWR-4 power board - Phase 2 leg A gate drive (S1, S2)

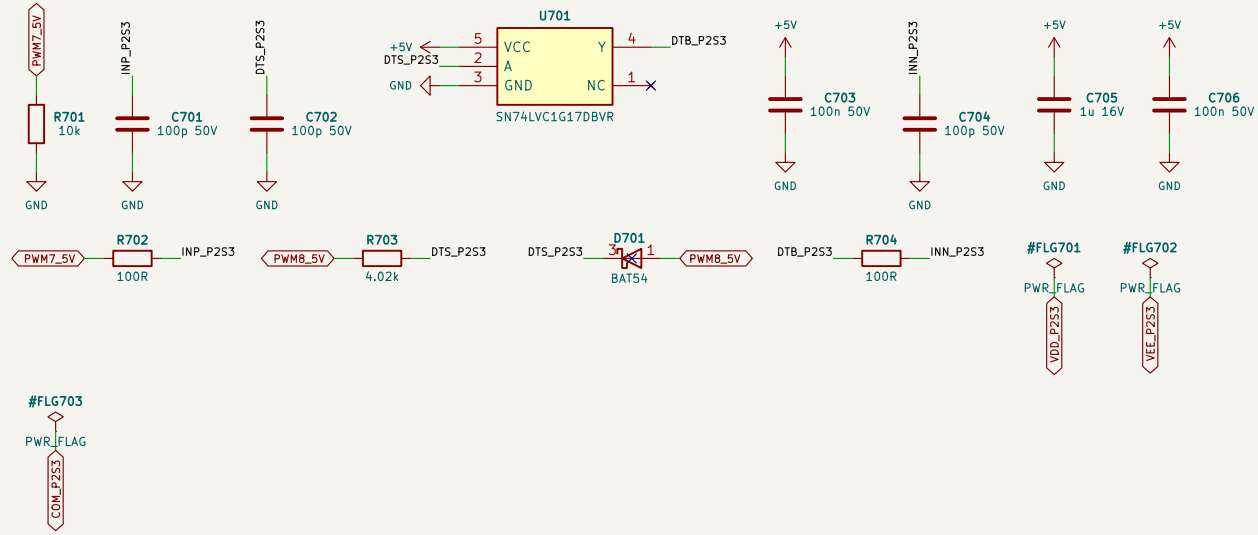
Size: A2	Date: 2026-10-05
----------	------------------

KiCad E.D.A. 10.0.6

Rev: A2

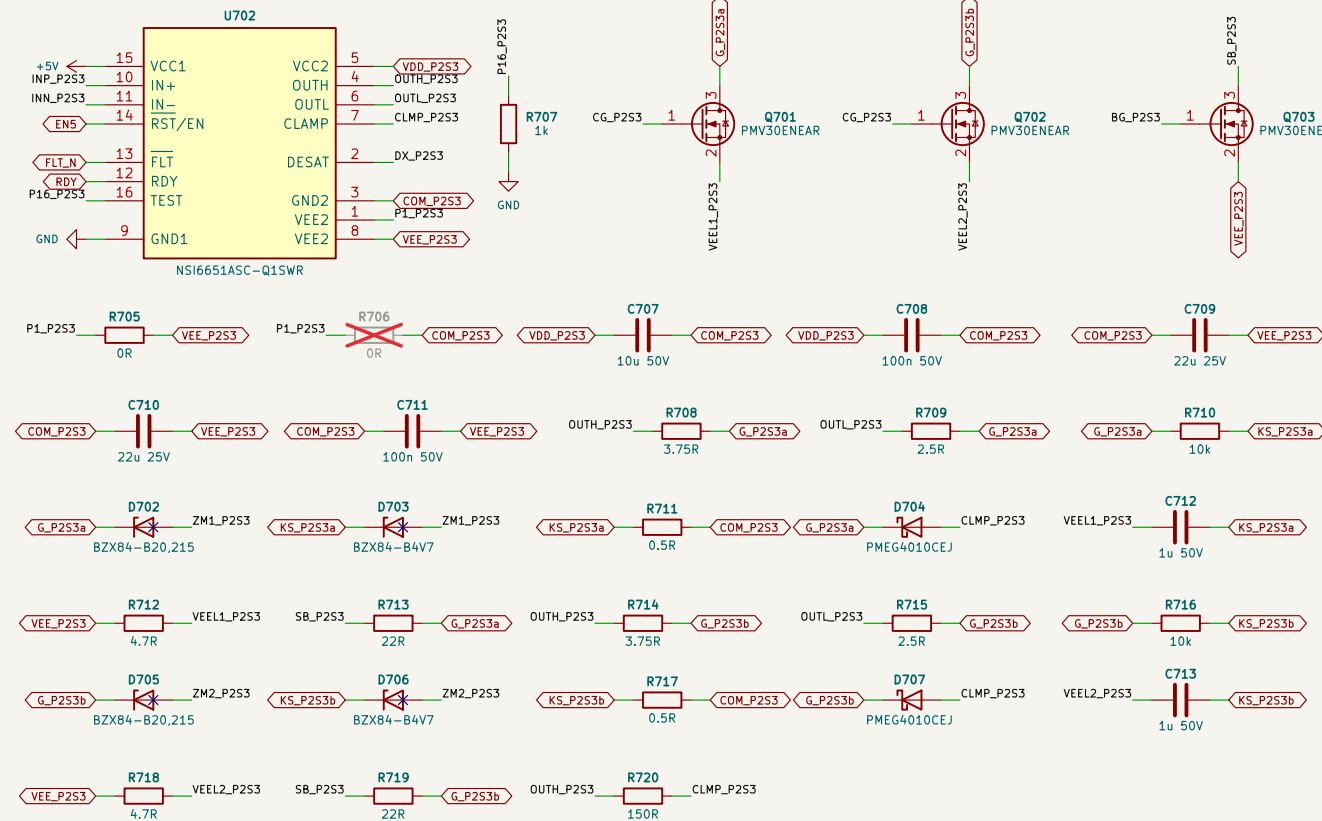
Id: 7/26

P2S3: LOGIC SIDE



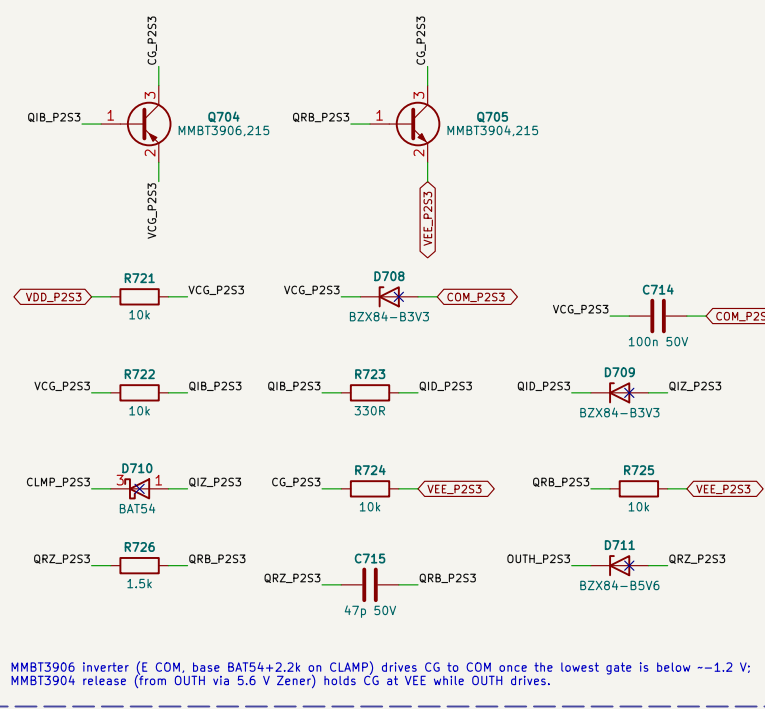
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



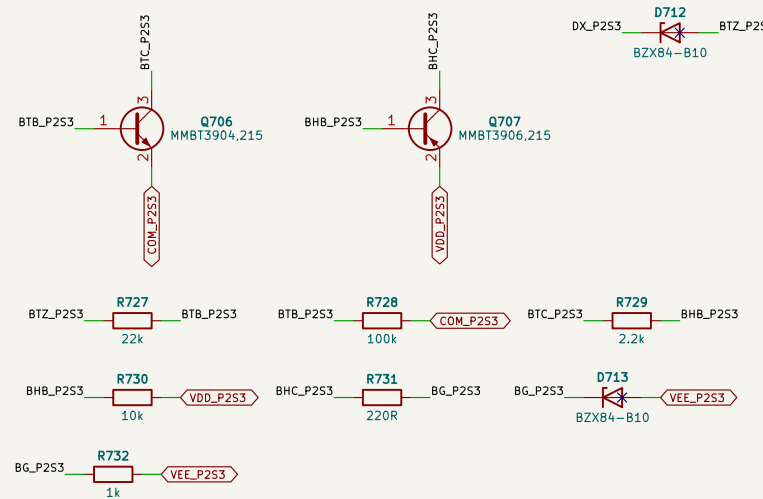
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL

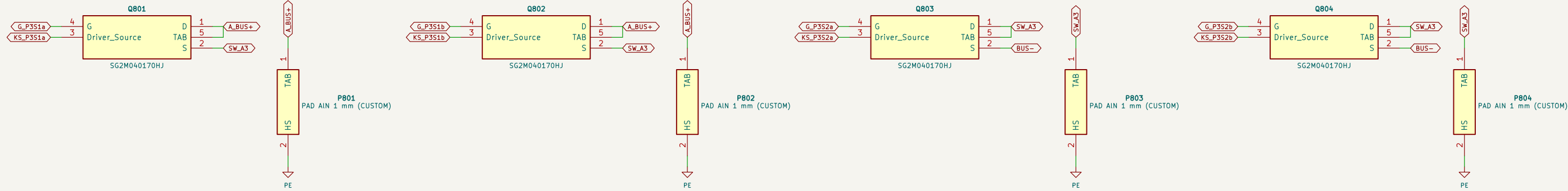


MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER

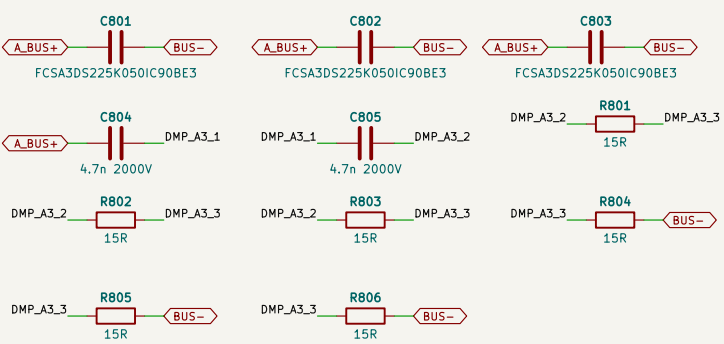


PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



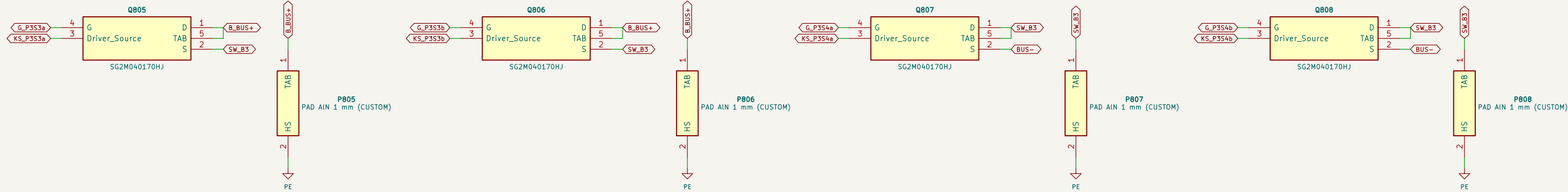
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



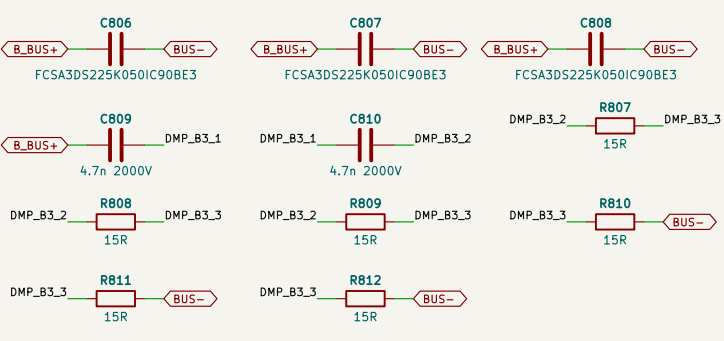
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



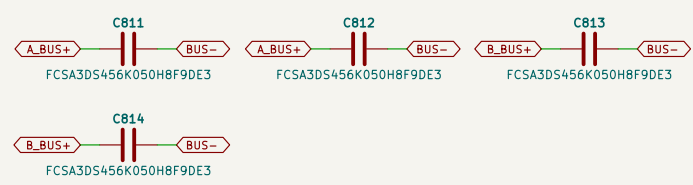
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



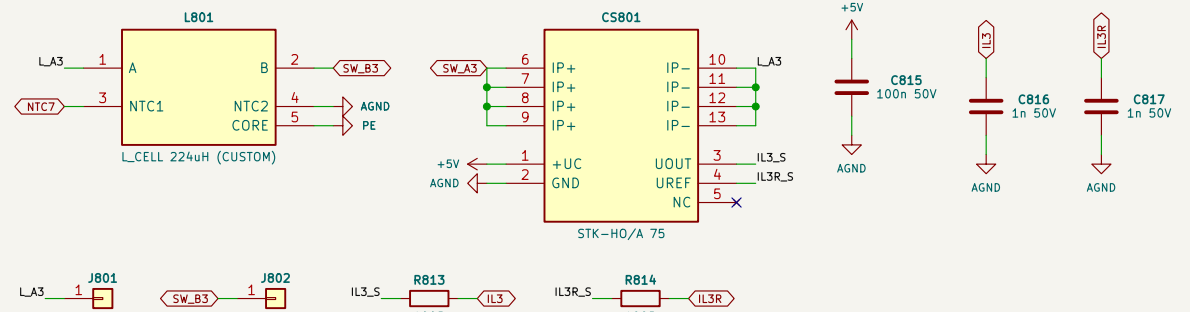
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



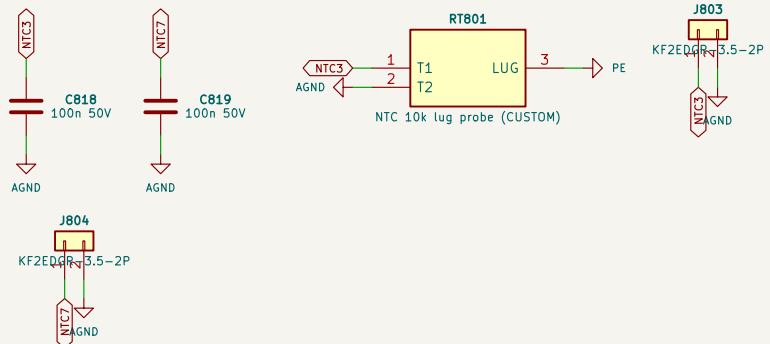
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



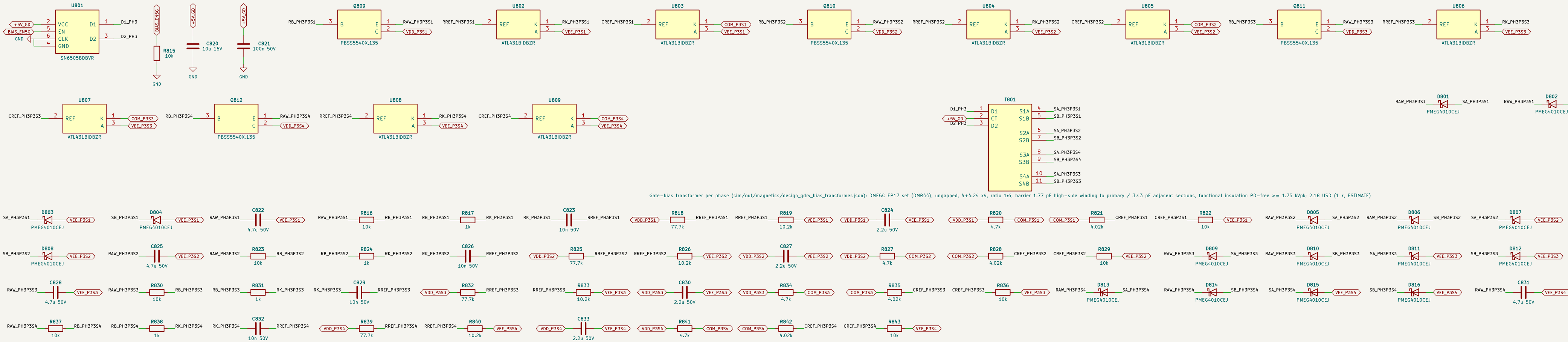
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out) + = SW_A -> SW_B) to stud
LA3; lead B on stud SW_B3. IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mV at the source. IL3R = the sensor's own Uref (pin 4). 100R +/- 1 mV; PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



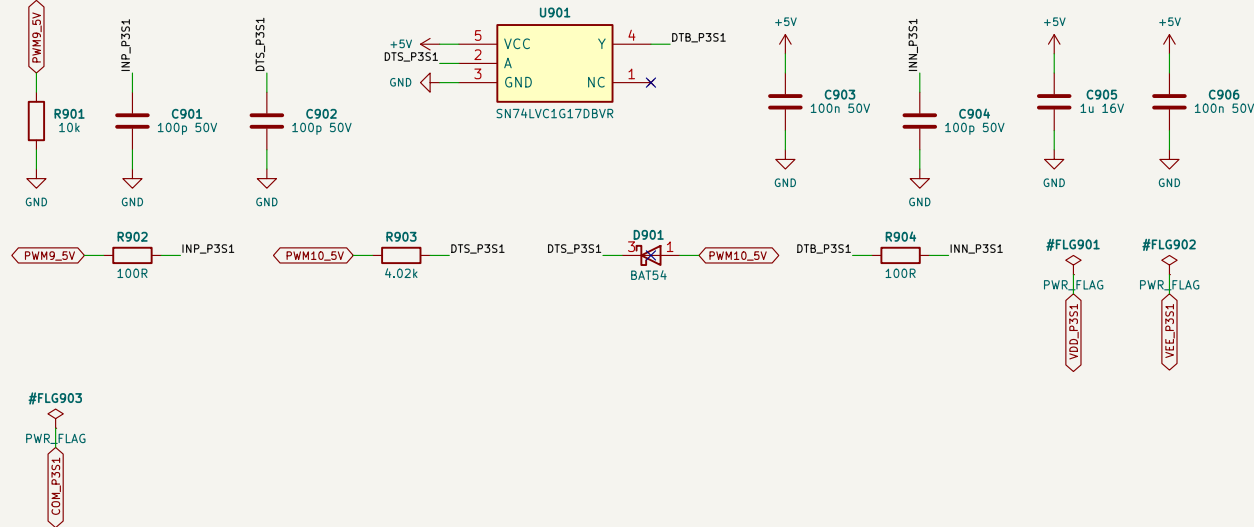
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC7 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



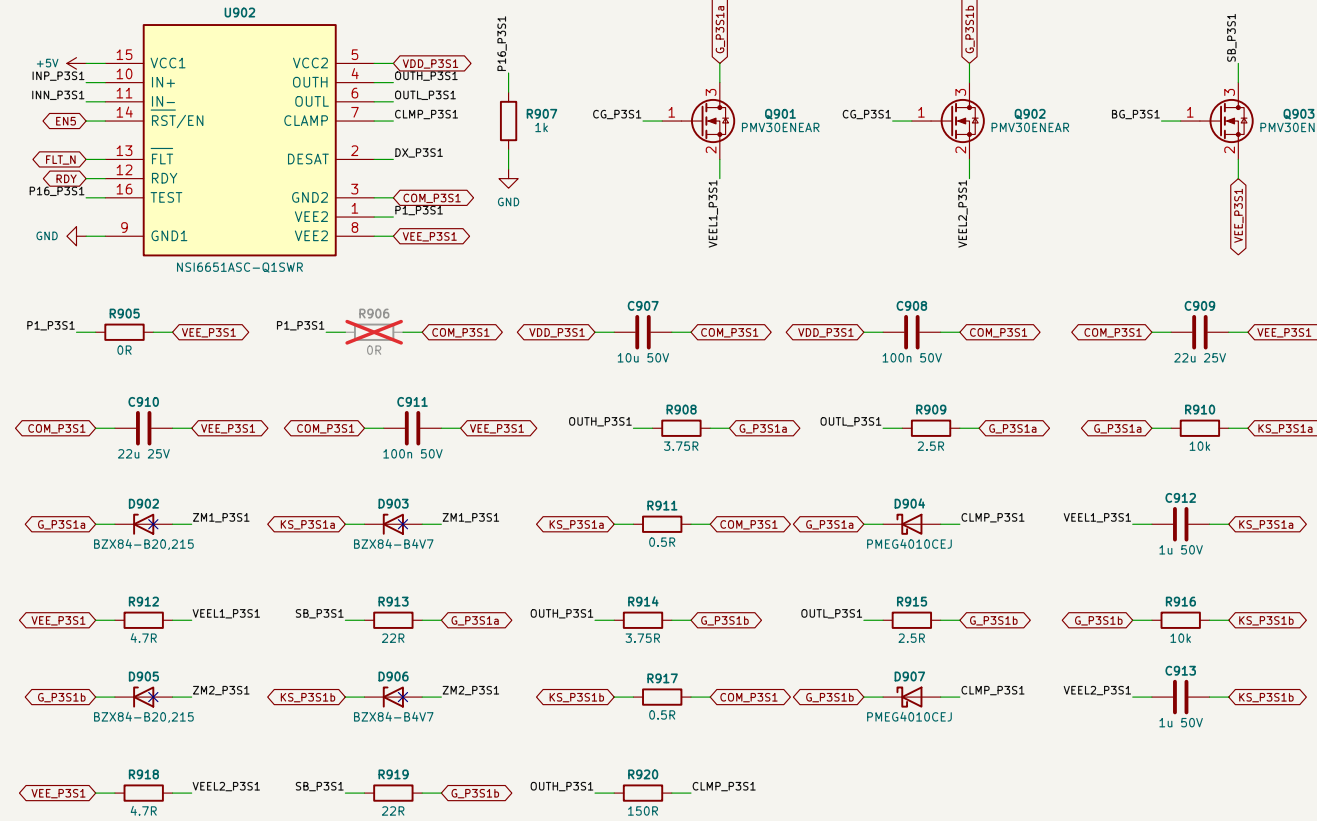
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



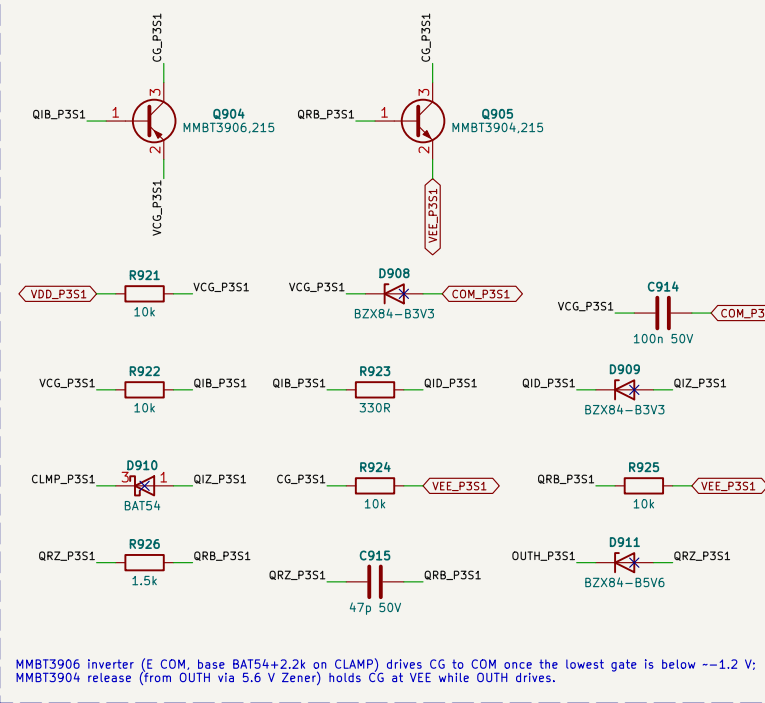
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



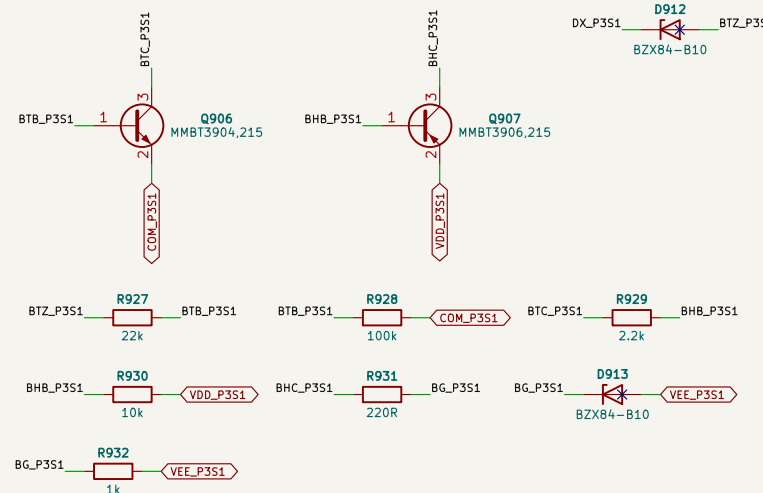
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



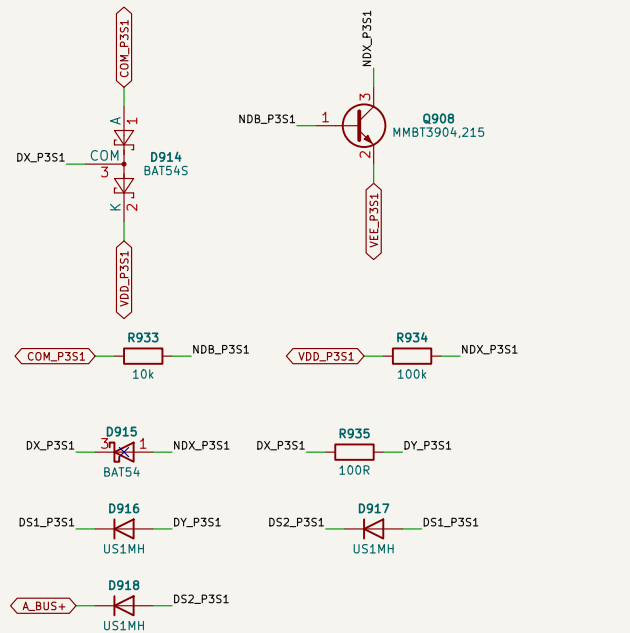
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



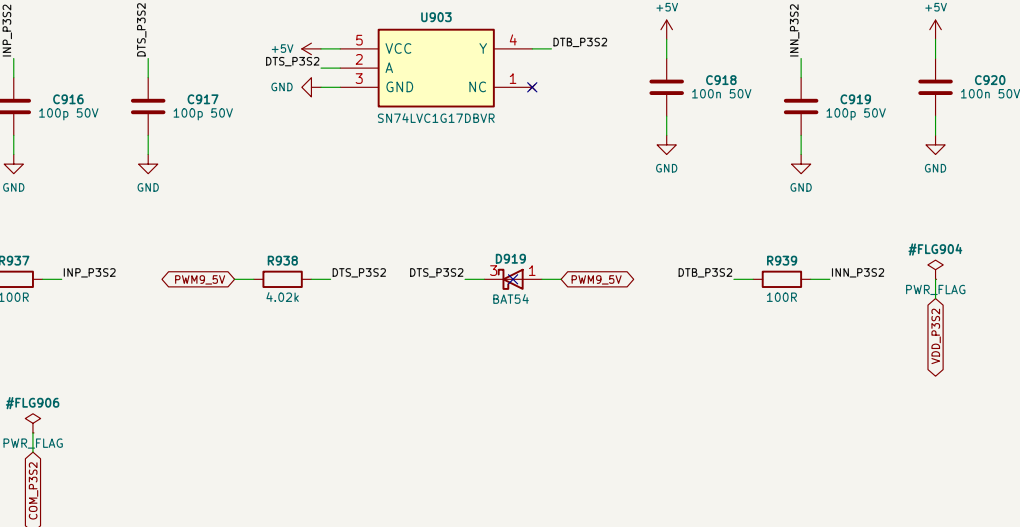
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P3S1: DESAT



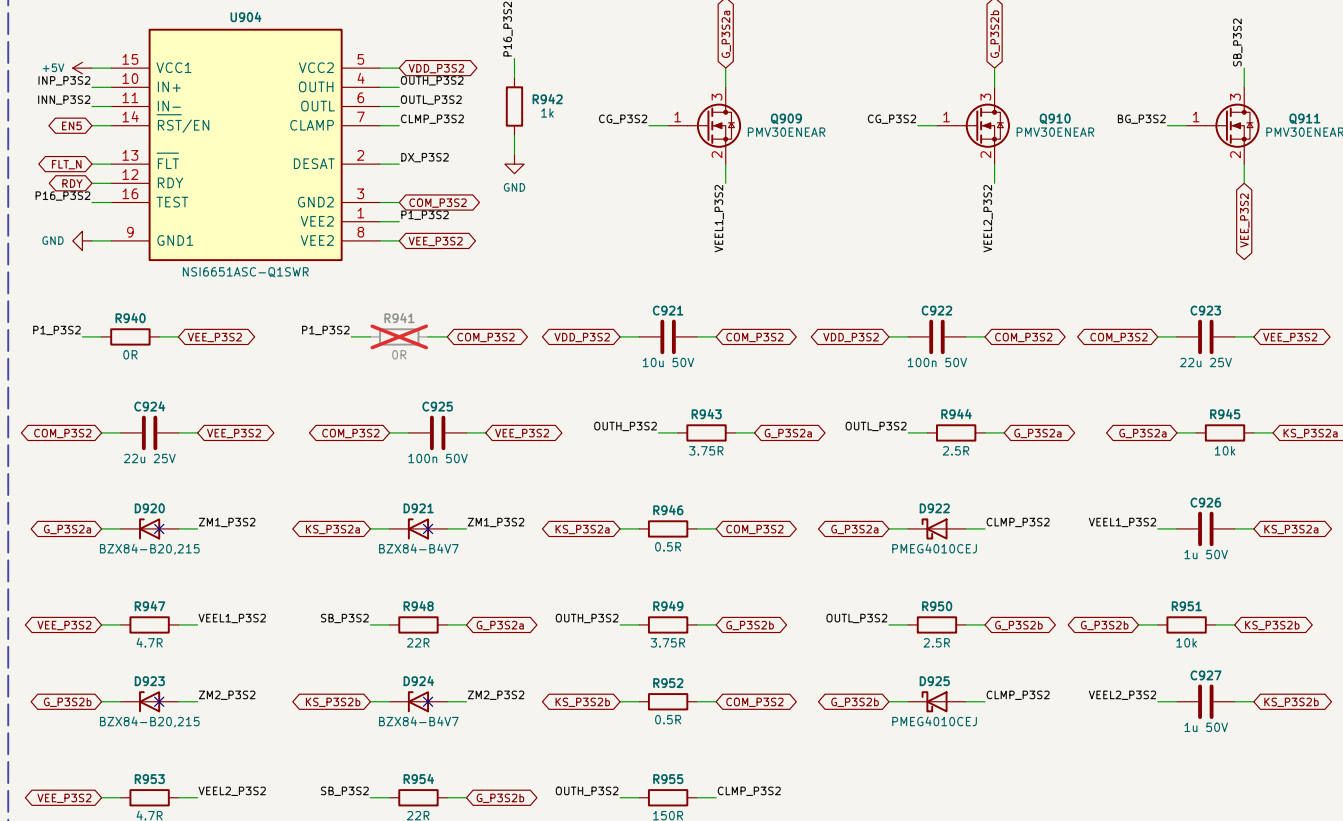
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbk footprint DNP.

P3S2: LOGIC SIDE



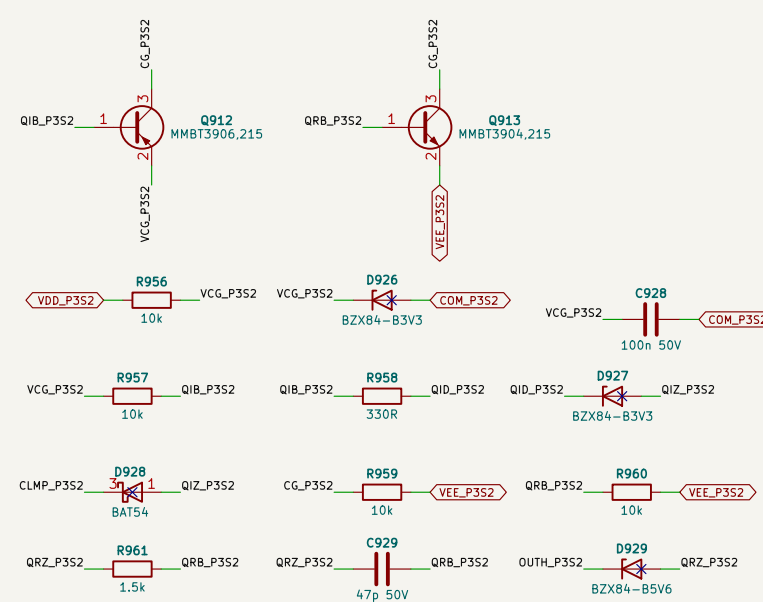
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



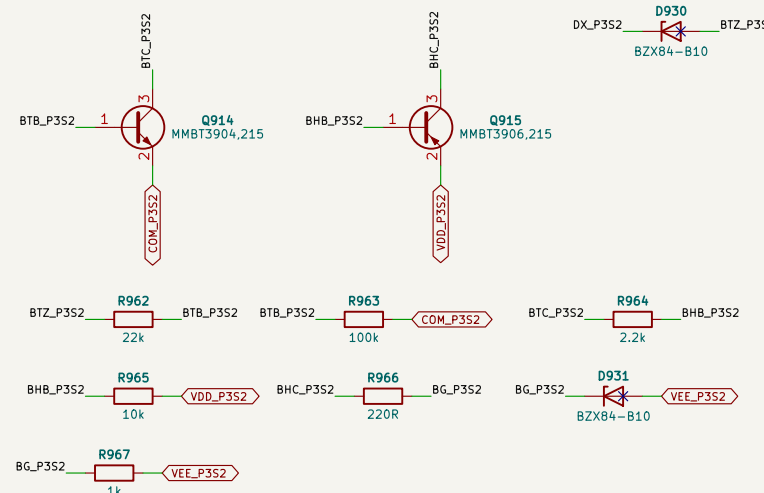
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



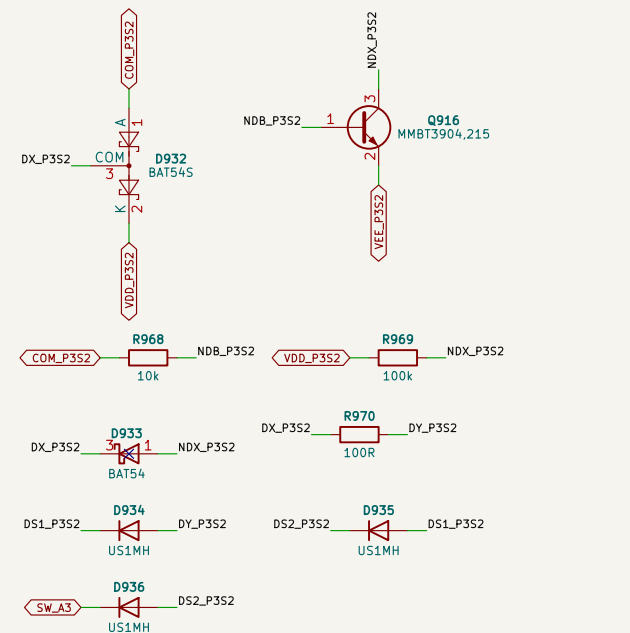
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (Lk + FET gates).

P3S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) -> Rk - 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbk footprint DNP.

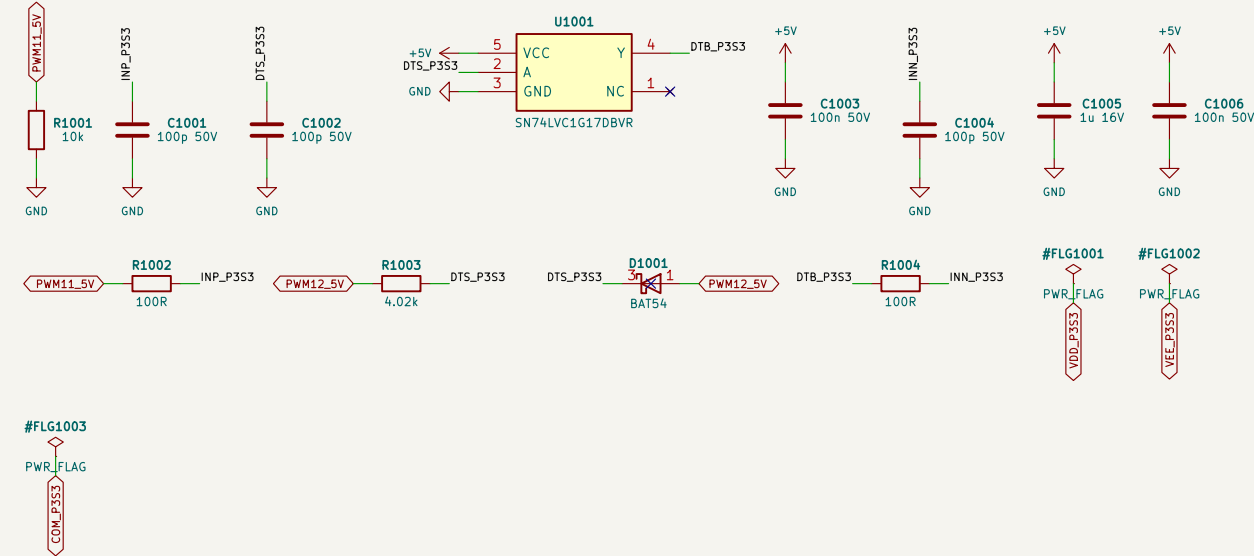
LEG A: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (highside, gnd); die +1.41 V vs VGS(on) min 1.94 V at 1/5 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 10
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

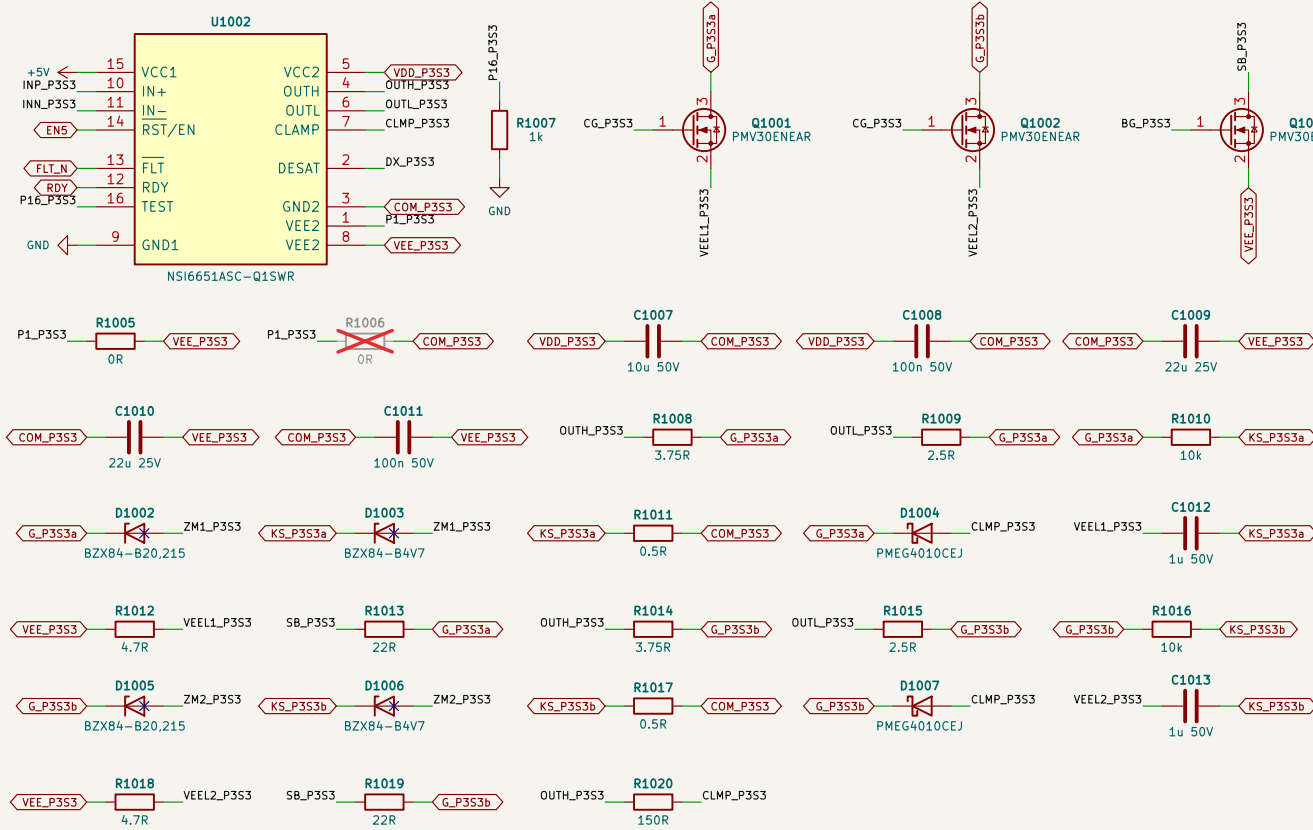
Sheet: /09 Phase 3 leg A gate drive (S1, S2)/
File: 09_ph3_gda.kicad_sch
Title: PV-PWR-4 power board - Phase 3 leg A gate drive (S1, S2)
Size: A2 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 10/26

P3S3: LOGIC SIDE



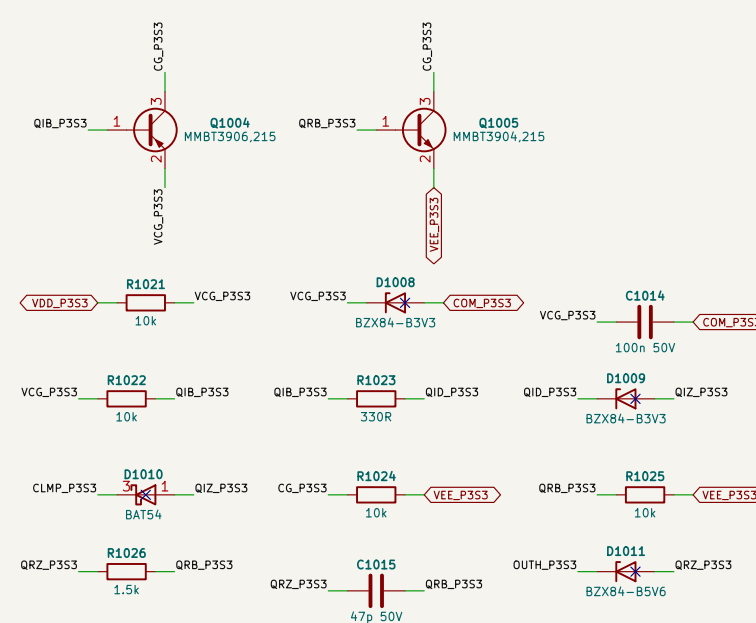
In+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



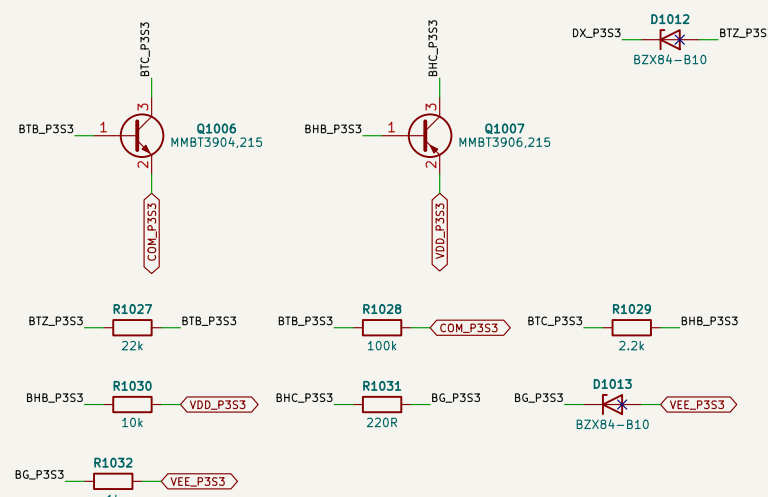
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



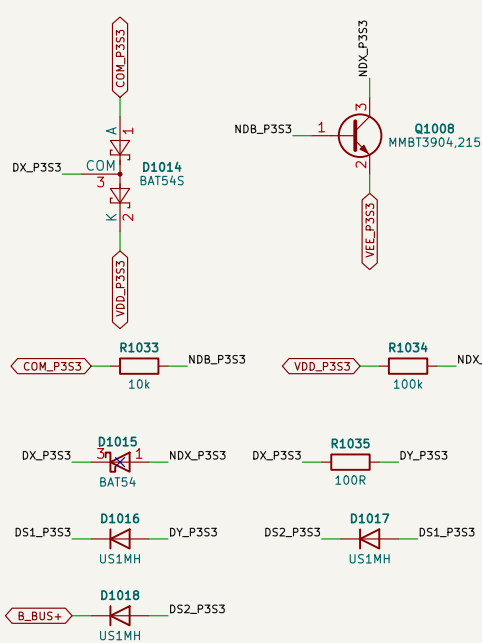
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



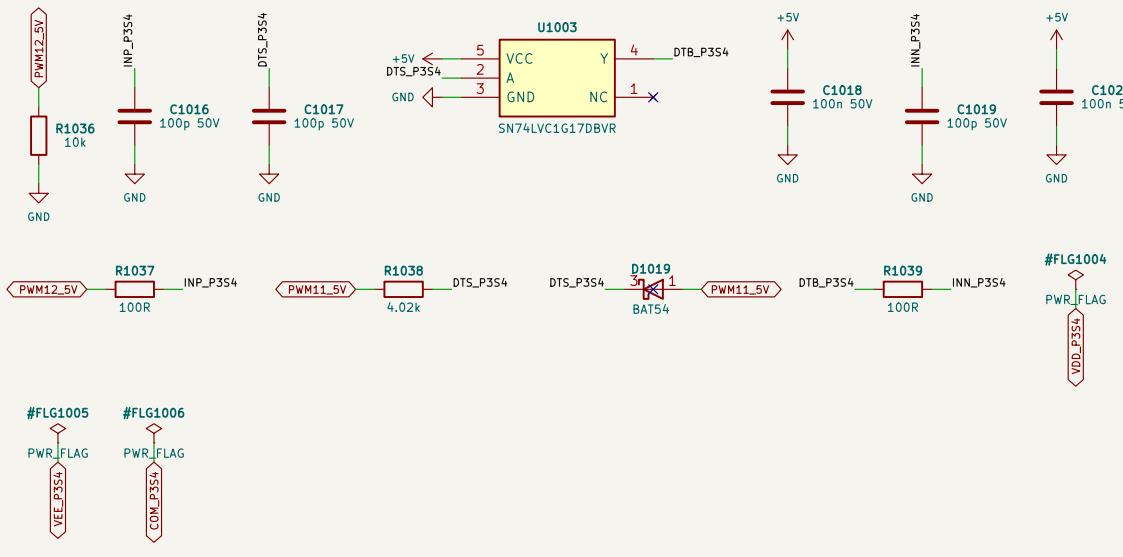
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S3: DESAT



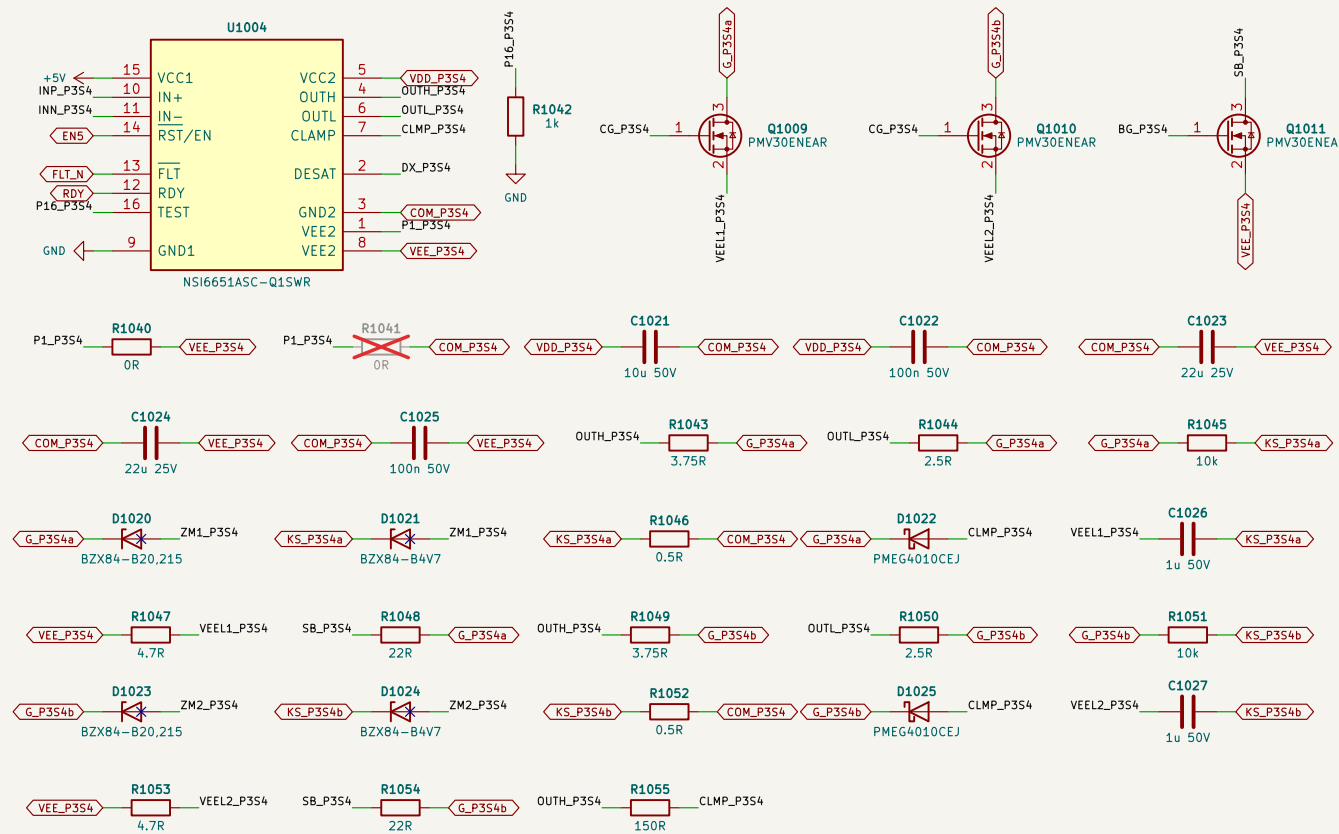
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S4: LOGIC SIDE



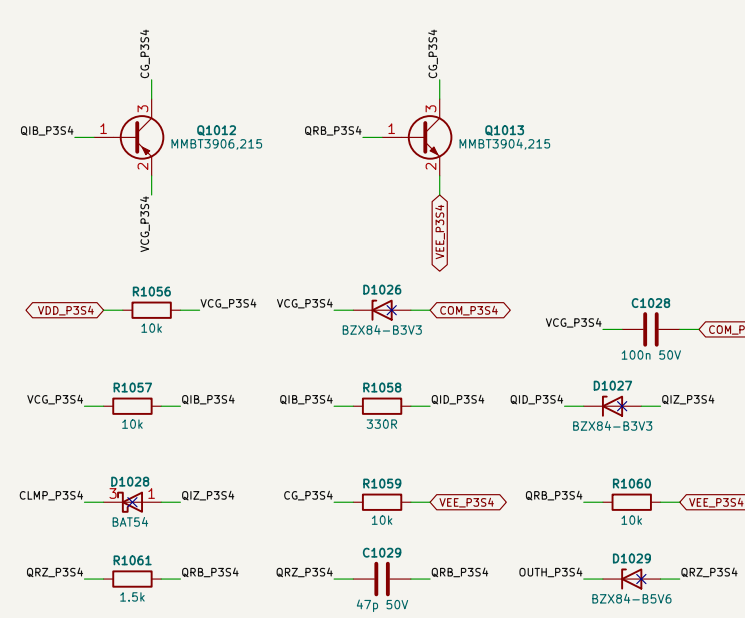
In+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



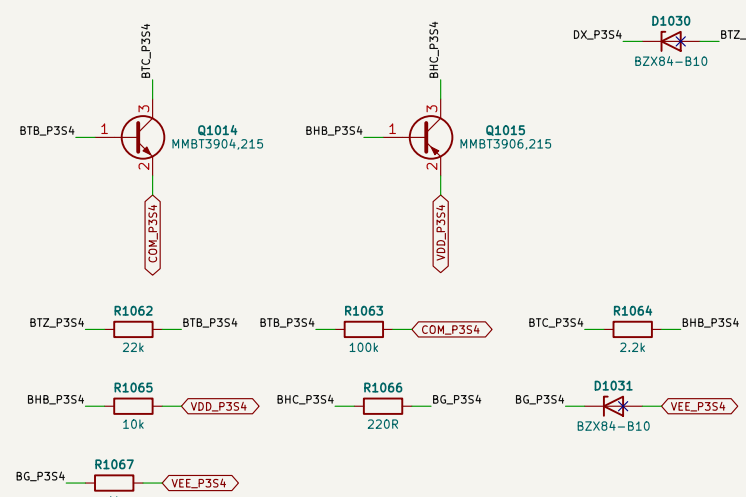
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



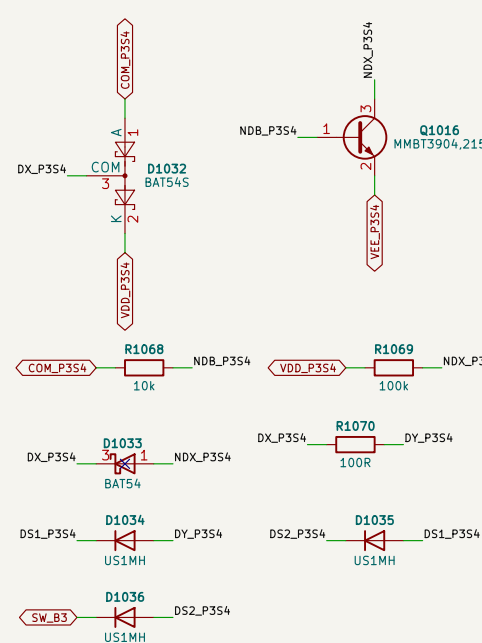
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (Lk + FET gates).

P3S4: DESAT

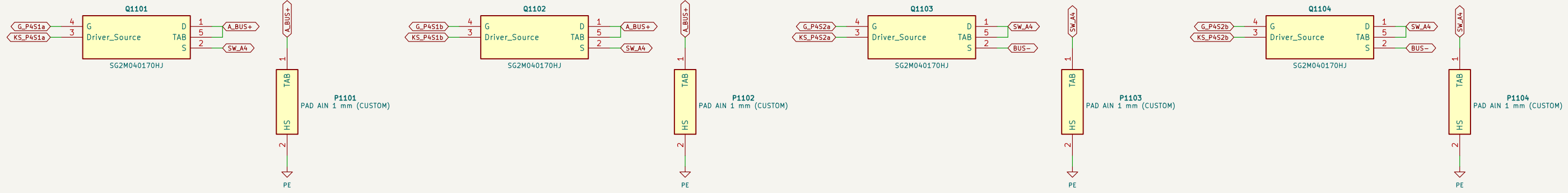


DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rk - 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

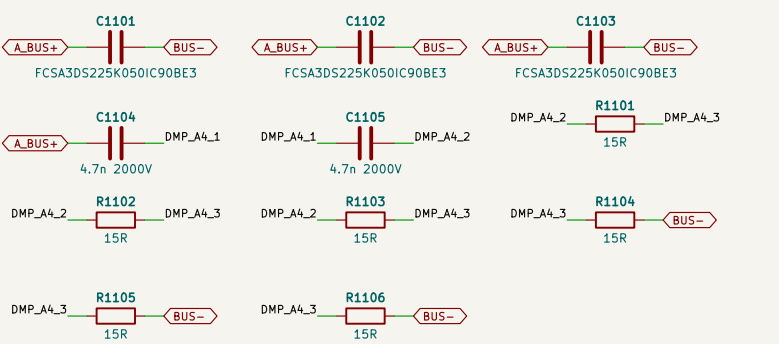
Gate and Kelvin of each device as coupled pairs of equal length (+/- 10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string l = 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (nigpice, gdrv); die +1.41 V vs VGS(on) min 1.94 V at 1/5. C = BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



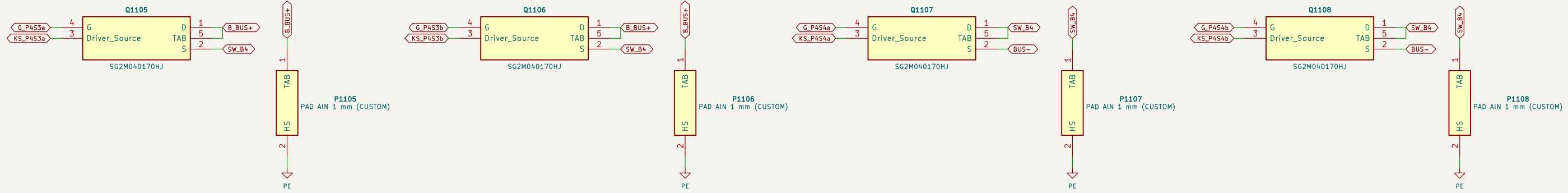
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



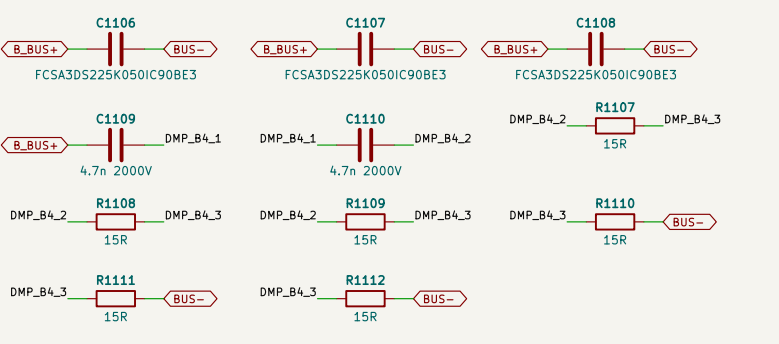
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



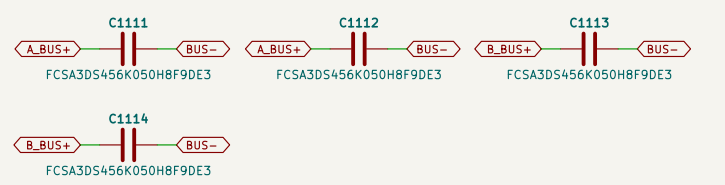
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



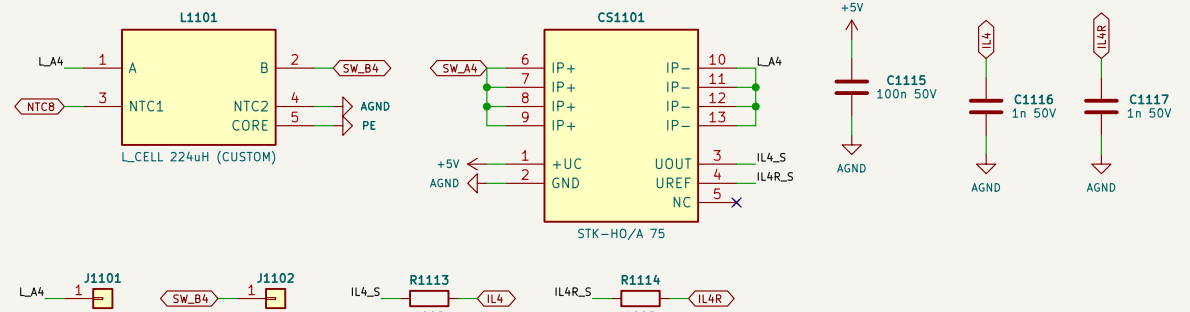
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



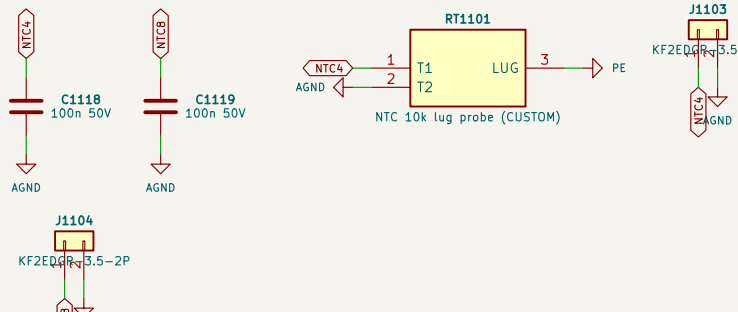
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



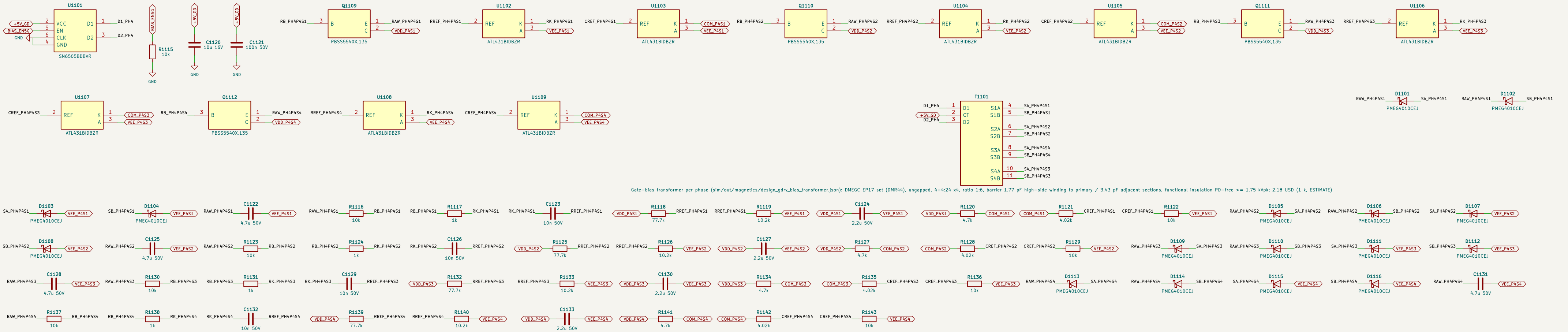
Lead A from SW_A4 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud
L_A4; lead B on stud SW_B4; IL4 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mV at the source; IL4R = the sensor's own Uref (pin 4). 100R +/- 1 mV; PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC4 = heatsink section of phase 4: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array I_{sc} ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TYT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up. eff 3793 V pole-P1 and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Sheet: /14 Port A power path (pv)/		
File: 14_leanA_power.kicad_sch		
Title: PV-PWR-4 power board - Port A power path (pv)		
Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 15/26

12345678

A

B

C

D

E

F

12345678

A

B

C

D

E

F

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Port A sensing and interlocks/
File: 15_leanA_ctrl.kicad_sch

Title: **PV-PWR-4 power board – Port A sensing and interlocks**

Size: A3Date: 2026-10-05

KiCad E.D.A. 10.0.6

Rev: **A2**
Id: 16/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/
File: 16_leanA_drive.kicad_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 17/26

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HPE501 250 A aR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HPE501 250 A aR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

[illegible]

The image displays a complex PCB layout for a 100W Class D amplifier. The layout is organized into several functional blocks, each containing specific components and their interconnections. Key components include:

- ICs:** U901 (SN74VC1G11BVR), U902 (SN74VC1G11BVR), U903 (SN74VC1G11BVR), U904 (SN74VC1G11BVR), U905 (SN74VC1G11BVR), U906 (SN74VC1G11BVR), U907 (SN74VC1G11BVR), U908 (SN74VC1G11BVR), U909 (SN74VC1G11BVR), U910 (SN74VC1G11BVR), U911 (SN74VC1G11BVR), U912 (SN74VC1G11BVR), U913 (SN74VC1G11BVR), U914 (SN74VC1G11BVR), U915 (SN74VC1G11BVR), U916 (SN74VC1G11BVR), U917 (SN74VC1G11BVR), U918 (SN74VC1G11BVR), U919 (SN74VC1G11BVR), U920 (SN74VC1G11BVR), U921 (SN74VC1G11BVR), U922 (SN74VC1G11BVR), U923 (SN74VC1G11BVR), U924 (SN74VC1G11BVR), U925 (SN74VC1G11BVR), U926 (SN74VC1G11BVR), U927 (SN74VC1G11BVR), U928 (SN74VC1G11BVR), U929 (SN74VC1G11BVR), U930 (SN74VC1G11BVR), U931 (SN74VC1G11BVR), U932 (SN74VC1G11BVR), U933 (SN74VC1G11BVR), U934 (SN74VC1G11BVR), U935 (SN74VC1G11BVR), U936 (SN74VC1G11BVR), U937 (SN74VC1G11BVR), U938 (SN74VC1G11BVR), U939 (SN74VC1G11BVR), U940 (SN74VC1G11BVR), U941 (SN74VC1G11BVR), U942 (SN74VC1G11BVR), U943 (SN74VC1G11BVR), U944 (SN74VC1G11BVR), U945 (SN74VC1G11BVR), U946 (SN74VC1G11BVR), U947 (SN74VC1G11BVR), U948 (SN74VC1G11BVR), U949 (SN74VC1G11BVR), U950 (SN74VC1G11BVR).
- Capacitors:** C1001 (100nF), C1002 (100nF), C1003 (100nF), C1004 (100nF), C1005 (100nF), C1006 (100nF), C1007 (100nF), C1008 (100nF), C1009 (100nF), C1010 (100nF), C1011 (100nF), C1012 (100nF), C1013 (100nF), C1014 (100nF), C1015 (100nF), C1016 (100nF), C1017 (100nF), C1018 (100nF), C1019 (100nF), C1020 (100nF), C1021 (100nF), C1022 (100nF), C1023 (100nF), C1024 (100nF), C1025 (100nF), C1026 (100nF), C1027 (100nF), C1028 (100nF), C1029 (100nF), C1030 (100nF), C1031 (100nF), C1032 (100nF), C1033 (100nF), C1034 (100nF), C1035 (100nF), C1036 (100nF), C1037 (100nF), C1038 (100nF), C1039 (100nF), C1040 (100nF), C1041 (100nF), C1042 (100nF), C1043 (100nF), C1044 (100nF), C1045 (100nF), C1046 (100nF), C1047 (100nF), C1048 (100nF), C1049 (100nF), C1050 (100nF), C1051 (100nF), C1052 (100nF), C1053 (100nF), C1054 (100nF), C1055 (100nF), C1056 (100nF), C1057 (100nF), C1058 (100nF), C1059 (100nF), C1060 (100nF), C1061 (100nF), C1062 (100nF), C1063 (100nF), C1064 (100nF), C1065 (100nF), C1066 (100nF), C1067 (100nF), C1068 (100nF), C1069 (100nF), C1070 (100nF), C1071 (100nF), C1072 (100nF), C1073 (100nF), C1074 (100nF), C1075 (100nF), C1076 (100nF), C1077 (100nF), C1078 (100nF), C1079 (100nF), C1080 (100nF), C1081 (100nF), C1082 (100nF), C1083 (100nF), C1084 (100nF), C1085 (100nF), C1086 (100nF), C1087 (100nF), C1088 (100nF), C1089 (100nF), C1090 (100nF), C1091 (100nF), C1092 (100nF), C1093 (100nF), C1094 (100nF), C1095 (100nF), C1096 (100nF), C1097 (100nF), C1098 (100nF), C1099 (100nF), C1100 (100nF), C1101 (100nF), C1102 (100nF), C1103 (100nF), C1104 (100nF), C1105 (100nF), C1106 (100nF), C1107 (100nF), C1108 (100nF), C1109 (100nF), C1110 (100nF), C1111 (100nF), C1112 (100nF), C1113 (100nF), C1114 (100nF), C1115 (100nF), C1116 (100nF), C1117 (100nF), C1118 (100nF), C1119 (100nF), C1120 (100nF), C1121 (100nF), C1122 (100nF), C1123 (100nF), C1124 (100nF), C1125 (100nF), C1126 (100nF), C1127 (100nF), C1128 (100nF), C1129 (100nF), C1130 (100nF), C1131 (100nF), C1132 (100nF), C1133 (100nF), C1134 (100nF), C1135 (100nF), C1136 (100nF), C1137 (100nF), C1138 (100nF), C1139 (100nF), C1140 (100nF), C1141 (100nF), C1142 (100nF), C1143 (100nF), C1144 (100nF), C1145 (100nF), C1146 (100nF), C1147 (100nF), C1148 (100nF), C1149 (100nF), C1150 (100nF), C1151 (100nF), C1152 (100nF), C1153 (100nF), C1154 (100nF), C1155 (100nF), C1156 (100nF), C1157 (100nF), C1158 (100nF), C1159 (100nF), C1160 (100nF), C1161 (100nF), C1162 (100nF), C1163 (100nF), C1164 (100nF), C1165 (100nF), C1166 (100nF), C1167 (100nF), C1168 (100nF), C1169 (100nF), C1170 (100nF), C1171 (100nF), C1172 (100nF), C1173 (100nF), C1174 (100nF), C1175 (100nF), C1176 (100nF), C1177 (100nF), C1178 (100nF), C1179 (100nF), C1180 (100nF), C1181 (100nF), C1182 (100nF), C1183 (100nF), C1184 (100nF), C1185 (100nF), C1186 (100nF), C1187 (100nF), C1188 (100nF), C1189 (100nF), C1190 (100nF), C1191 (100nF), C1192 (100nF), C1193 (100nF), C1194 (100nF), C1195 (100nF), C1196 (100nF), C1197 (100nF), C1198 (100nF), C1199 (100nF), C1200 (100nF), C1201 (100nF), C1202 (100nF), C1203 (100nF), C1204 (100nF), C1205 (100nF), C1206 (100nF), C1207 (100nF), C1208 (100nF), C1209 (100nF), C1210 (100nF), C1211 (100nF), C1212 (100nF), C1213 (100nF), C1214 (100nF), C1215 (100nF), C1216 (100nF), C1217 (100nF), C1218 (100nF), C1219 (100nF), C1220 (100nF), C1221 (100nF), C1222 (100nF), C1223 (100nF), C1224 (100nF), C1225 (100nF), C1226 (100nF), C1227 (100nF), C1228 (100nF), C1229 (100nF), C1230 (100nF), C1231 (100nF), C1232 (100nF), C1233 (100nF), C1234 (100nF), C1235 (100nF), C1236 (100nF), C1237 (100nF), C1238 (100nF), C1239 (100nF), C1240 (100nF), C1241 (100nF), C1242 (100nF), C1243 (100nF), C1244 (100nF), C1245 (100nF), C1246 (100nF), C1247 (100nF), C1248 (100nF), C1249 (100nF), C1250 (100nF), C1251 (100nF), C1252 (100nF), C1253 (100nF), C1254 (100nF), C1255 (100nF), C1256 (100nF), C1257 (100nF), C1258 (100nF), C1259 (100nF), C1260 (100nF), C1261 (100nF), C1262 (100nF), C1263 (100nF), C1264 (100nF), C1265 (100n

HP6501/0008100-250 in both poles: breaks 1.25-50 kA only (partial range);
the part QC trip and the contactor keep it out of 250 A-1.25 kA.

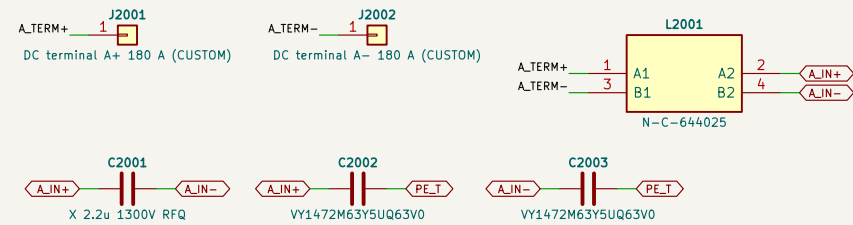
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F	1	2	3	4	5	6	7	8

Not bench-validated. Values read from sim/out/*.json; checks: design_check()		
Sheet 19		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /17 Port B sensing and interlocks/		
File: 17_leanB_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 19/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

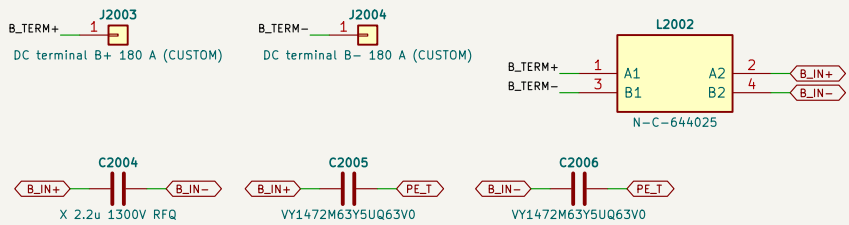
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 20 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /18 Port B coil drives/ File: 18_leanB_drive.kicad_sch		
Title: PV-PWR-4 power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 20/26	

PORT A TERMINALS AND EMI FILTER



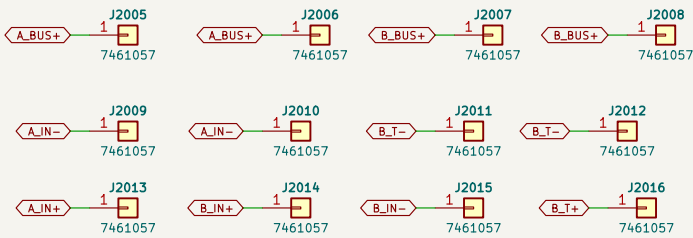
PV: chassis terminals A_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



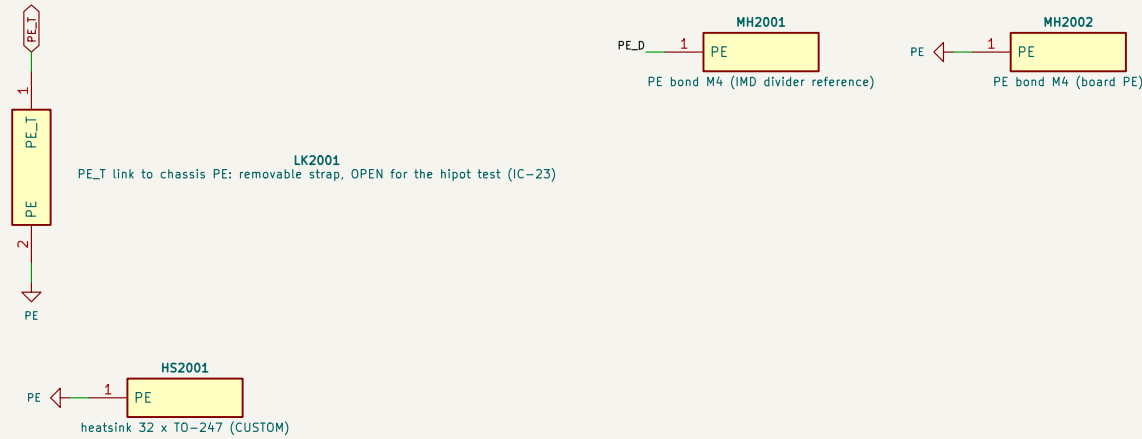
battery: chassis terminals B_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



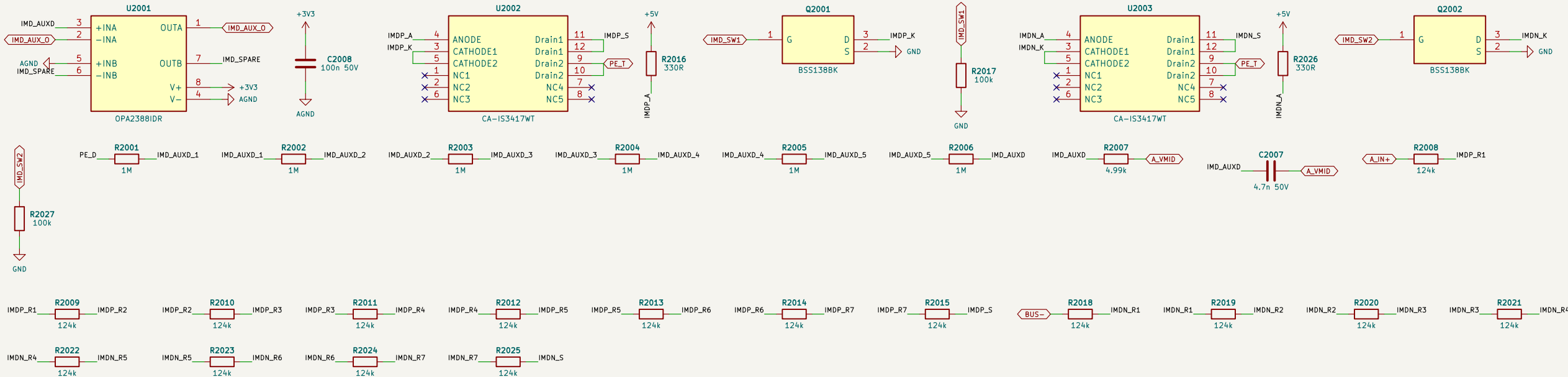
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



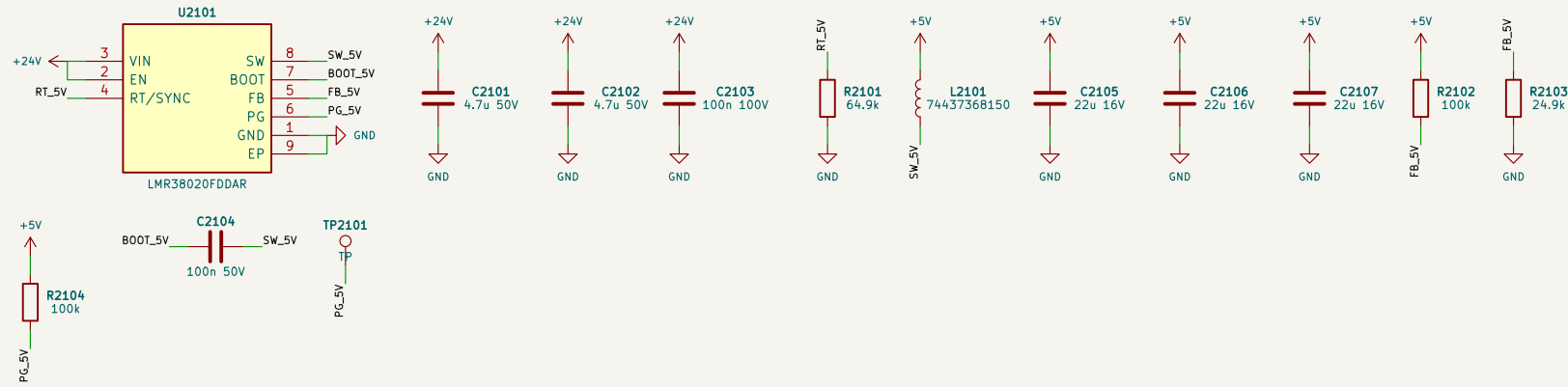
PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



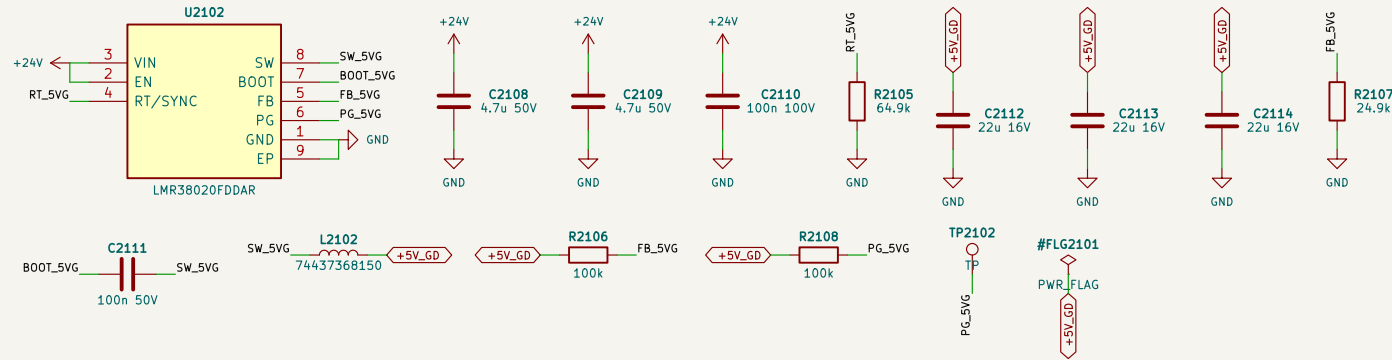
IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



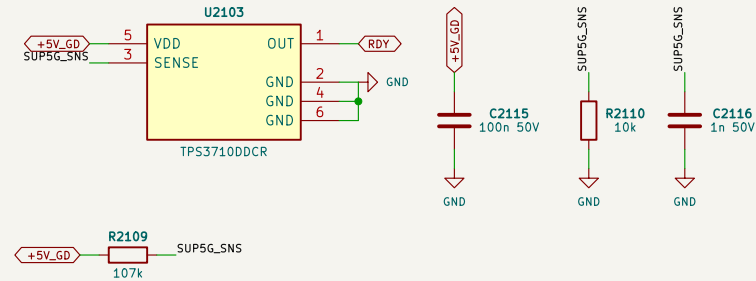
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1–2, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



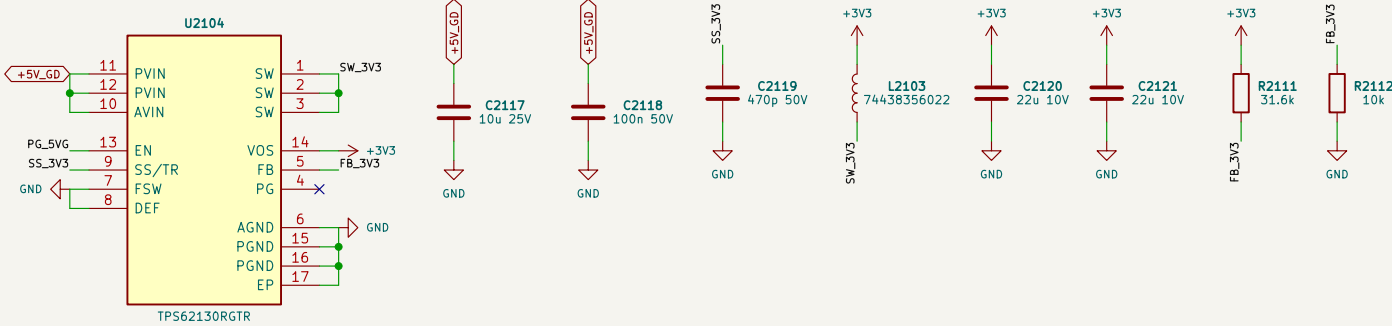
gate bias of phases 3–4, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



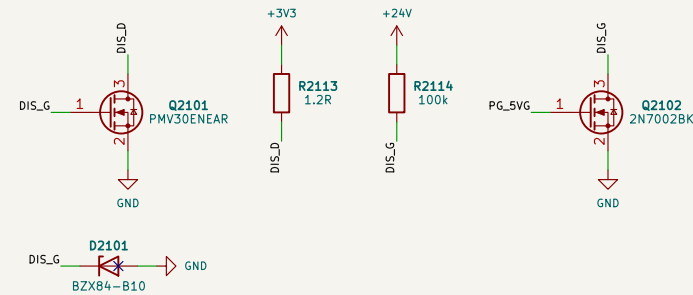
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3–4 is below 4.52–4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240–3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12–16 mV/us (F28003x VDDIO 8–100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90–94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX–T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 21.4–30.1 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 22

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

Sheet: /19 Live 5 V / 3.3 V, SELV connector/

File: 19_supplies.kicad_sch

Title: PV–PWR–4 power board – Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026–10–05

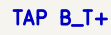
Rev: A2

KiCad E.D.A. 10.0.6

Id: 22/26

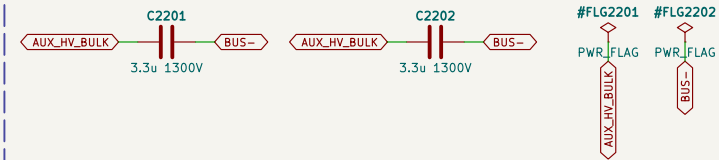


2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting



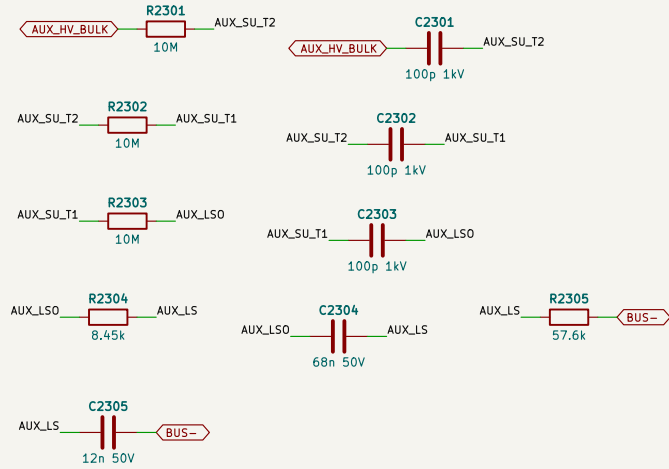
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



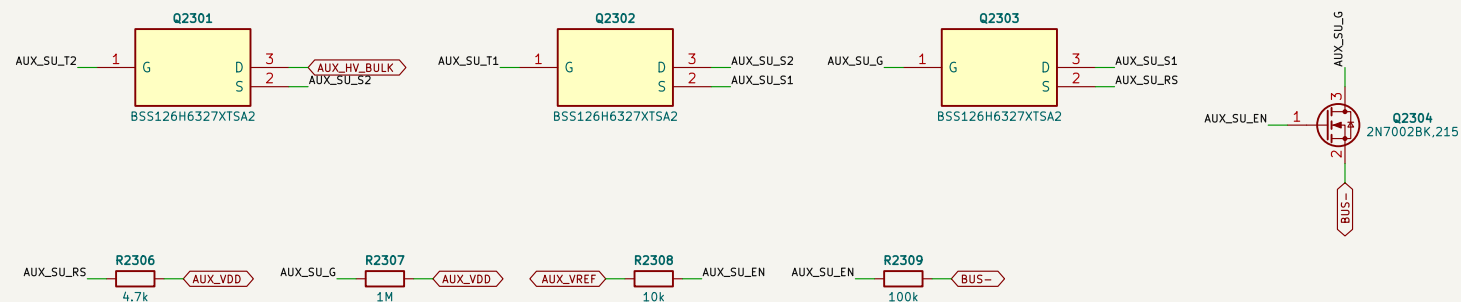
Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

COMPENSATED TAP STRING



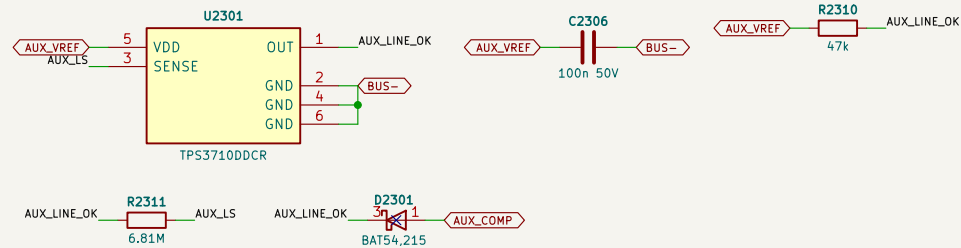
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



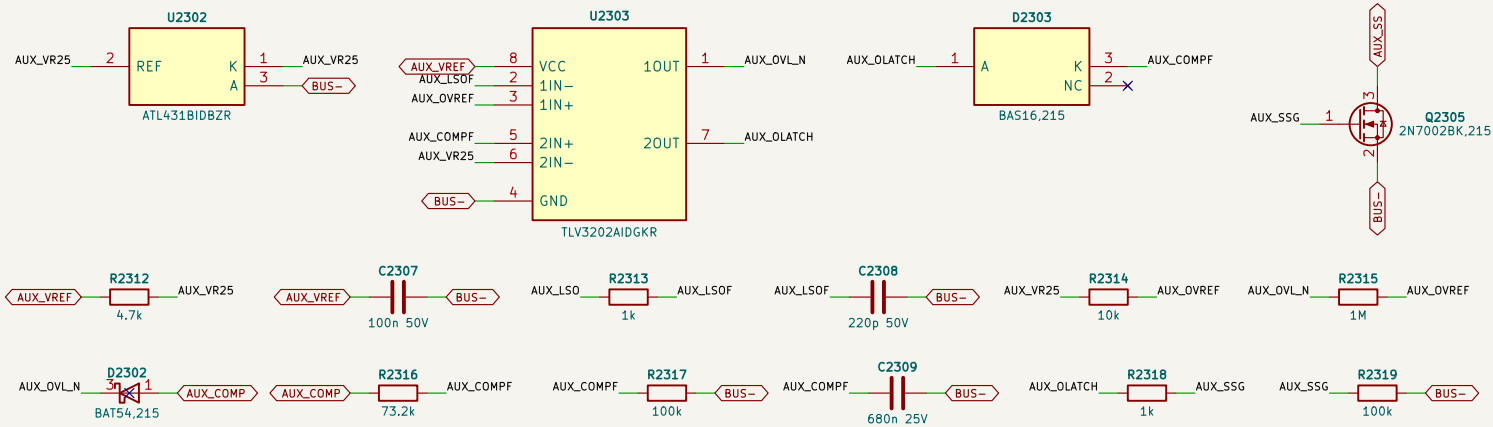
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



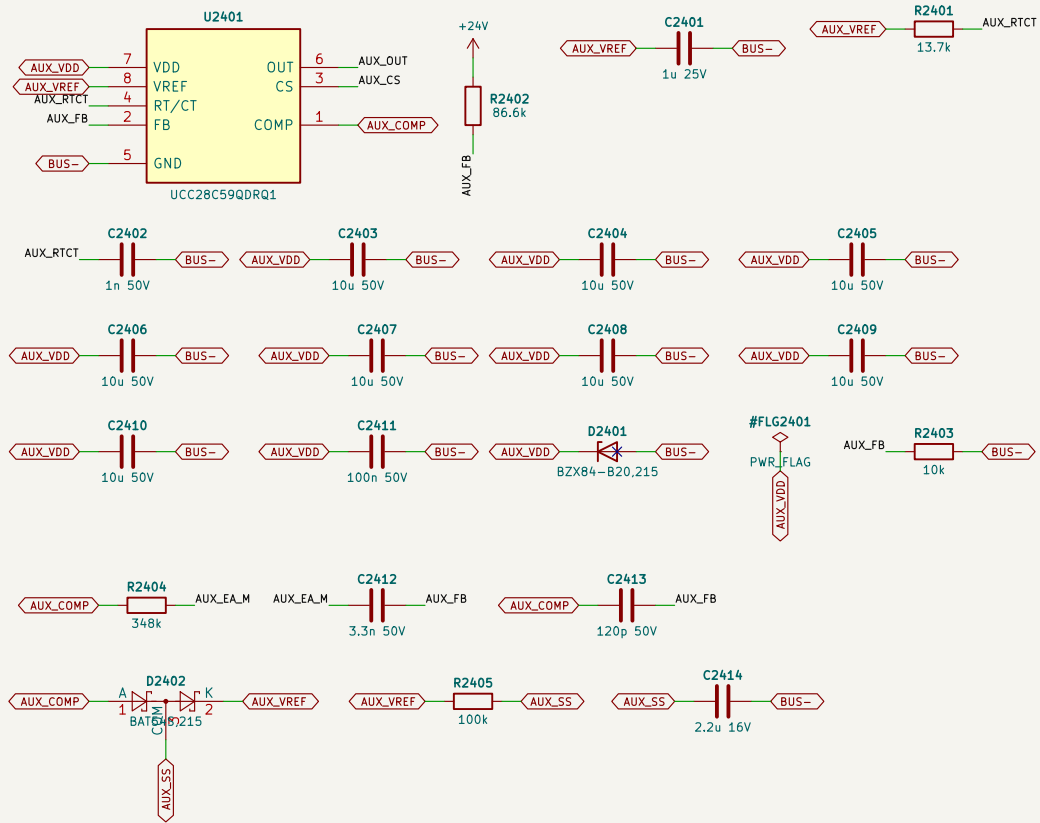
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



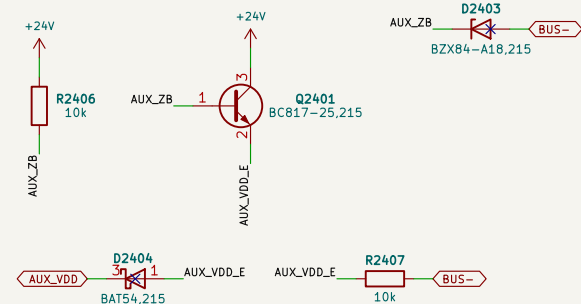
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms -> latch -> SS crowbar -> hiccup.

PWM CONTROLLER AND ERROR AMPLIFIER



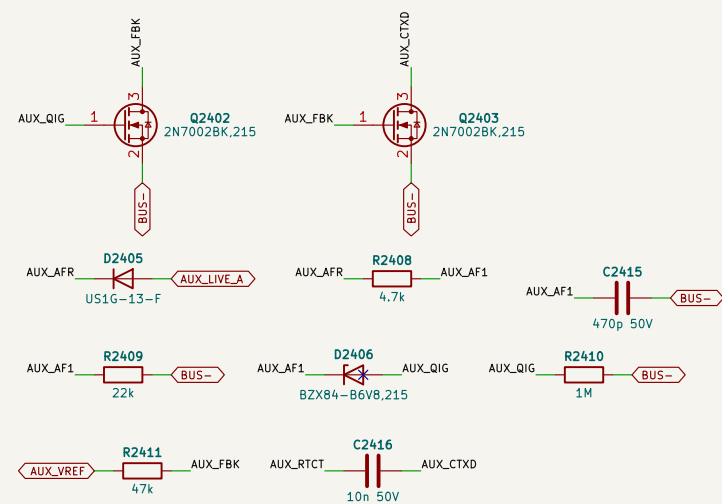
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 259–727 Hz, phase margin >= 60 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



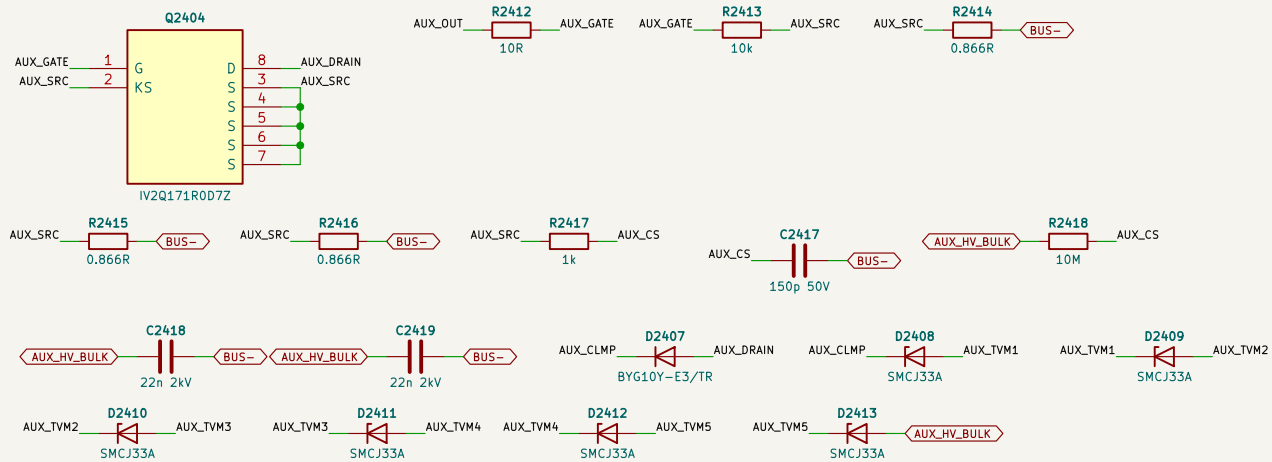
(30 W case, unchanged) follower Zener bias: VDD = gate drive 15.46–18.11 V (BZX84–A18, –40..85 C)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX–03)



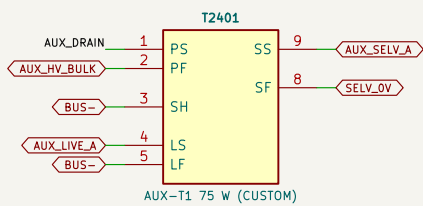
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output -> fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 in parallel = 0.288667R (E96); lowest limit 3.04 A delivers 112.8 W (>= the 112.2 W 1 s peak) at every input; highest 4.07 A with the line feed-forward (4.46 A without)
line feed-forward HV_BULK -> CS (CRHV2512); CS offset 101 mV at 1000 V cancels most of the 155–250 ns CS-delay overshoot, so the highest current limit no longer rises with the input
Clamp: BYG10Y + 6 x SMCJ33A; V_DS 1318 / 1422 V at 1000 / 1100 V.

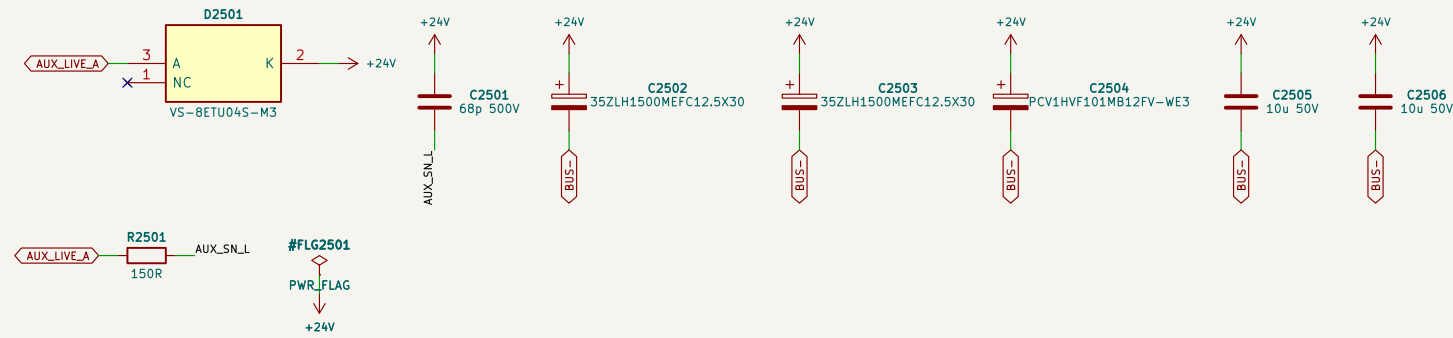
AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/- 7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

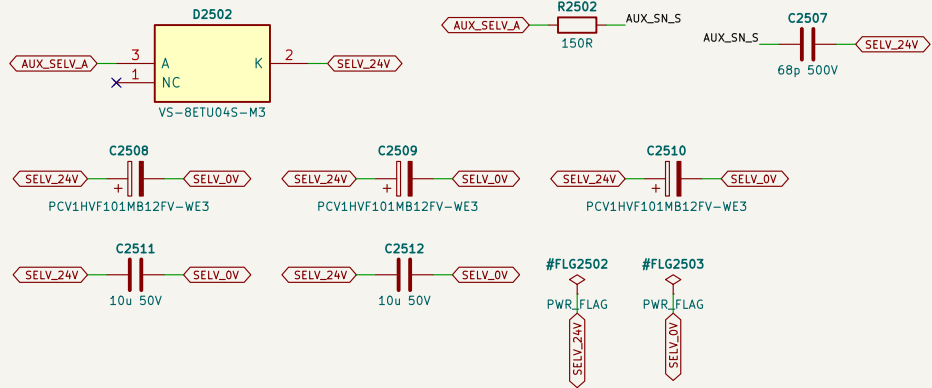
Not bench-validated. Values read from sim/out/*.json; checks: design_check()		
Sheet 25		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /22 AUX75 PWM, SiC, clamp, AUX-T1/ File: 22_aux75_power.kicad_sch		
Title: PV-PWR-4 power board – AUX75 PWM, SiC, clamp, AUX-T1		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 25/26

LIVE 24 V RECTIFIER, HOLD-UP



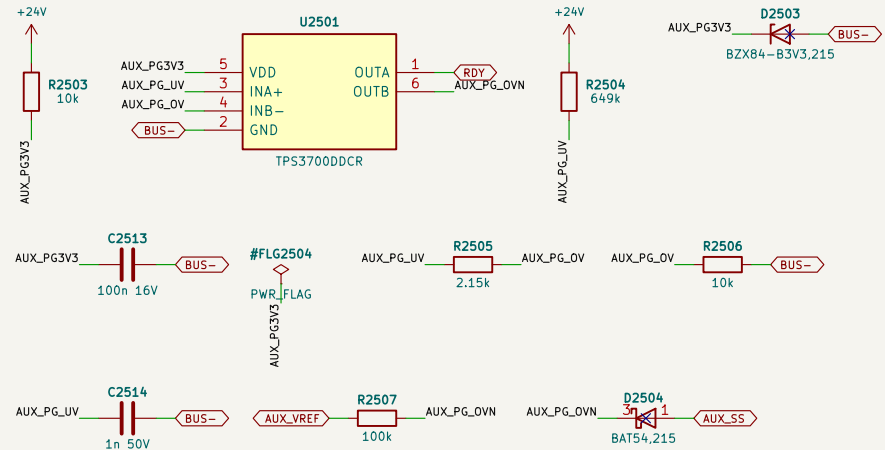
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)